QUASAR

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FEATURED ARTICLES

Mobile Apps – To CE Mark and Beyond

Data Display Considerations for Medical Device Data Systems

Electronic Technology – Introducing Safe Havens – The Grampian Way!



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RQA Research Quality Association

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QUASAR

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KATIE WILDE



Summary Comparison of Global GLP Regulations Pertaining to Liquid Chromatography – Mass Spectrometry Instrument Software Validation

DAVE ABRAMOWITZ, JING MA AND STACY NELSON





The Use of Relational Modelling as a Tool to Detect Safety Signals Amongst Patients' Web Posts

JAMES SAWYER

Veterinary Quality Assurance and Computer Systems Validation

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Electronic Source Documentation – Current Regulatory Expectations and Challenges for Industry

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Editor's Note



This edition of Quasar focuses on 'Electronic Technology' and has yielded eight topical articles. Thanks to all the contributing authors:

'Mobile Apps – To CE Mark and Beyond' by Rhona McAteer and Kevin Bambury discusses how unprofessional, unapproved medical apps are coming to an end and how the industry has to begin embracing the new age of regulation, resulting in approved apps.

'Data Display Considerations for Medical Device Data Systems' by Fraser Smith gives an insight into the FDA final rule for an electronic medical device, covering topics such as real time data processing, irreversible data compression, numerical vs. graphical displays, along with recommendations.

'Summary Comparison of Global GLP Regulations Pertaining to Liquid Chromatography – Mass Spectrometry Instrument Software Validation' by Dave Abramowitz, Jing Ma and Stacy Nelson gives an overview of the similarities and differences between global regulations. The full article can be found in the new online articles area on the RQA website.

'The Use of Relational Modelling as a Tool to Detect Safety Signals Amongst Patients' Web Posts' by James Sawyer looks at how the healthcare industry has an obligation to evaluate all data sources; those within its control and those potentially beyond.

'A Spreadsheet Validation Case Study' by Thana Subramanian uses the scenario of un-validated spreadsheets being used in a laboratory, the applicable regulations and guidance, basic spreadsheets, control and simplification, validation approach, risk assessment, requirements specification, configuration and code review, through to testing and reporting.

'Electronic Technology – Introducing Safe Havens – The Grampian Way!' by Katie Wilde focuses on how to maintain high levels of security and good governance processes to protect patient confidentiality, whilst at the same time facilitating high quality data linkage research studies.

'Electronic Source Documentation– Current Regulatory Expectations and Challenges for Industry' by Jon Read provides an analysis of the FDA draft guidance on Electronic Source Data in Clinical Investigations released in November 2012.

'Veterinary Quality Assurance and Computer Systems Validation' by Iain McPhee reviews the regulations and guidance, computer systems validation and the role of QA pre and post validation.

Also in this edition:

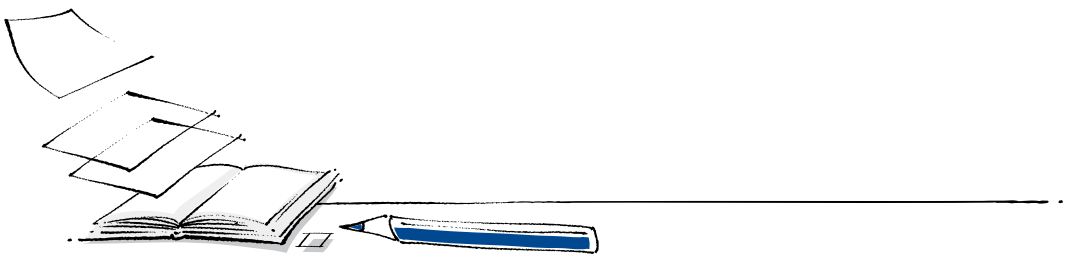
- Nichola Stevens' board biography as Chairman of the Computing Committee
- Viewpoint gives a view on 'QA Programmes and Service Provider Assessment' which discusses recent contaminated UK supermarket meat
- Updates from the Committees and Regional Forums.

Victoria

Articles on the Website

Why not visit the newly launched online articles area on the RQA website:

www.therqa.com/publications/onlinearticles



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Chairman's Message



LOUISE HANDY

Dear Friends,

Welcome to the spring issue of Quasar; yet again I find myself travelling as I write this. On this occasion I am heading to the fourth country I will have been to in five days and am already lining up more travel for later in the year. One reason for this is the **European QA Conference** which will be in Bonn from 25th – 27th September. We have already put out a call for posters and the exhibitor presence is starting to shape up nicely. You should shortly receive information for delegates, but please do check out the dedicated European QA Conference website at www.european-qa-conference.com or see the advert in this edition of Quasar for more details. I do hope that as many of you as possible will be able to support this unique initiative.

With regard to other activities, the RQA publication list has recently increased with the launch of a new computing booklet. **Auditing Computerised Systems** is a great addition to our membership resources and as so many aspects of our work now rely on this technology, we hope it will prove to be a valuable tool for you. We are also pleased to announce that the second phase of the **Regulatory Roadmap** is under way. This project will cover medical devices and is being developed in conjunction with the recently formed Medical Devices Committee. Additionally we launched our first webinar on the RQA website on the MHRA GCP Guide 2012. This follows on from a successful seminar on the topic and the **webinar** allows individuals to learn from a discussion of what elements of the GCP Guide appear to be new or different from the perspective of a GCP auditor. The added bonus is that this webinar is not restricted to a particular timeslot so you can access it at your leisure.

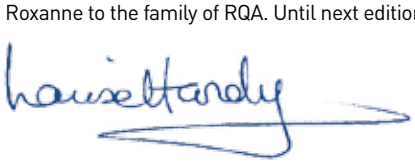
The RQA Board has been busy since the conference in November last year and one of the strands being reviewed is **marketing and products**. A small sub group of the Board, along with Tony Ward (our Development, Events and Publications Manager), has been reviewing our existing products and services, talking to 'users' and identifying gaps and opportunities for new products. We will be working on this further in line with our strategy so please do let us have any of your ideas for tools, courses or events that you would find useful.

As a Board we continuously consider how best to manage the Association's finances and at this time of year we normally make a general call to the membership for **proposals regarding investment concepts** which we can then consider for implementation in 2014. This could be enhancements to an existing service/area, or a proposal for a new service/facility, which also ties in with the work we are doing on marketing and products. Previous investments, for example, have included the development of eLearning courses and the Regulatory Roadmap, so please do contact us with your ideas.

Another Board activity has been to ensure that the **RQA Annual General Meeting (AGM)** is held in an appropriate time and place this year. It has been agreed that it will be held on 6th November 2013 at the National Motorcycle Museum, Solihull, West Midlands. The AGM will be held in conjunction with several individual seminars and you can learn more about this event on page 34 of this edition of Quasar. We hope you will all find something of interest and will be able to join us for the AGM.

I would also like to thank those of you who took the time to respond to the **Membership Satisfaction Survey**. We do take note of your responses and monitor our performance so your input is much appreciated. There will be feedback once we have had the opportunity to review this year's responses.

And finally I would like to welcome a new member of staff to the RQA office, Roxanne Holland, who will be providing cover for Danielle while she is on maternity leave. We wish Danielle all the best as she enters the realms of parenthood and welcome Roxanne to the family of RQA. Until next edition, best wishes



The 2013 cohort of the MSc in Quality Research and Development are working towards their thesis. Some of them require input from members, mainly in the form of filling in questionnaires. Please look at the Education and Training Committee news area of the website for information on each project and how you can get involved. Thank you.

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Board Biography

NICHOLA STEVENS



When I was growing up my career aspirations ranged from being a jockey (I was always too tall for that to be a viable option) to becoming an air traffic controller – a family friend even managed to arrange a trip to the control tower at Heathrow where I watched Concord being brought in to land. Somewhere along the way I became interested in biology and chemistry and ended up doing a degree in biochemistry. That led to my first job working for SmithKline & French performing small scale purifications of proteins. I became very used to working in the cold, the delightful smell of mercaptoethanol and coaxing the fast protein liquid chromatography (FPLC) to work.

After SmithKline and Beechams (SB) merged I transferred from The Frythe to Harlow and spent several years doing small molecule chromatography; I became a bit of specialist in chiral chromatography and a real advocate of the Chiralcel/Chiralpak range of high performance liquid chromatography (HPLC) columns. This was also my first introduction of working in a regulated environment as the whole department worked to GMP standards. Eventually I took the plunge and left SB to move to a small company called Oxford Asymmetry where I was able to broaden my knowledge of a variety of analytical chemistry techniques supporting manufacturing scale up activities. It was also my first introduction to computer system validation (CSV) when we implemented a new chromatography data system. Not only was the concept of CSV new and challenging to me but there was also all the concerns around the year 2000 and the impact that was going to have on all our computer systems. I clearly remember winding computer clocks forward to close to midnight 1999 (and a variety of other dates that were believed to be similarly dangerous) to see what would happen – such an anticlimax when everything carried on as before!

After too short a time working for Oxford Asymmetry my partner David announced that he had been offered his dream job at UMIST (which later became part of Manchester University) and so we relocated to Yorkshire. I joined Covance and validated another chromatography data system before changing job again to become a Validation Consultant working for Mi Services. I loved working as a consultant, having the opportunity to work with many different people in Europe and the US, seeing similar but different approaches to the same problems and learning so much along the way. Eventually however, I decided that I had had enough of living out of a suitcase and that it wasn't really fair to expect David to look after all my animals while I was away so I joined AstraZeneca and that is where I am today. I now work in the compliance function within corporate IS as the GxP Subject Matter Expert.

I joined the Computing Committee in early 2010 and became the Committee Chairman in November 2012.

I've already mentioned my animals; they take up quite a bit of my time outside of work, what with keeping the chickens out of the house, helping the cats get out of the house past the marauding chickens and exercising my horse when I can actually catch her! I also love horse racing (I'm a member at Cheltenham), mountain walking, climbing via ferrata and in recent years I've discovered skiing.



Mobile Apps – To CE Mark and Beyond

RHONA MCATEER AND KEVIN BAMBURY

At the turn of the 20th Century, the pharmaceutical industry was an unregulated free-for-all. You could put any product on the market and make whatever claims you wanted.



Remember the snake oil salesmen in the black and white cowboy movies? Products claimed to cure everything from cancer to corns, there were anti-fat remedies to make women slimmer, cocaine tooth-drops for tooth-ache and morphine for teething children. Fortunes were made from those who were desperately seeking a cure for their ailments, often with completely ineffective or at times dangerous drugs.

However, regulators soon stepped in and began controlling the products being put on the market and the claims being made. The wild-west style attitude to drugs disappeared and the modern 'pharma' industry we have today emerged. The pharmaceutical industry (despite its faults) has flourished and it produces drugs that are for the most part safe and effective, a big change from the madness of the turn of the 20th Century.

A similar turning point has also been reached with medical mobile applications (apps). A simple search of the appstore will show that there are apps on the market that claim to help fight depression and clear acne using your mobile phone, e.g. there is an app that uses blue light from the phone and must be rested on the affected area for 2 minutes on a daily basis. It costs $1.99 and has been downloaded 11,600 times. There is no literature cited to back the claims it is making. Numerous other apps are also available that offer cures from everything from shingles to insomnia, again the claims are mainly unapproved.

> They often panic at the question 'is your app a medical device?' responding with 'what is a medical device?'.

Even more frightening are the apps available to be used by doctors when making clinical decisions. Yes, the doctor stands between the app and the ultimate care decision, but doctors are human and they can make mistakes like everyone else. They may not double check every calculation an app produces. A simple programming error could have irreversible repercussions. Many apps used to calculate complex formulae that make life or death decisions have been created by software developers working outside the traditional medical device industry, often in small, spin-out companies or within universities or hospitals. These developers frequently have little or no experience of the regulatory environment and do not even realise their products may be subject to the Medical Device Directive (MDD). They often panic at the question 'is your app a medical device?' responding with 'what is a medical device?'.

Within the MDD, a medical device is defined as:

*'any instrument, apparatus, appliance, **software**, material or other article, whether used alone or in combination, including the software intended by its manufacturer to be used specifically for diagnostic and/or therapeutic purposes and necessary for its proper application, intended by the manufacturer to be used for human beings for the purpose of:*

- *diagnosis, prevention, monitoring, treatment or alleviation of disease*
- *diagnosis, monitoring, treatment, alleviation of or compensation for an injury or handicap*
- *investigation, replacement or modification of the anatomy or of a physiological process*
- *control of conception*

and which does not achieve its principal intended action in or on the human body by pharmacological, immunological or metabolic means, but which may be assisted in its function by such means'.

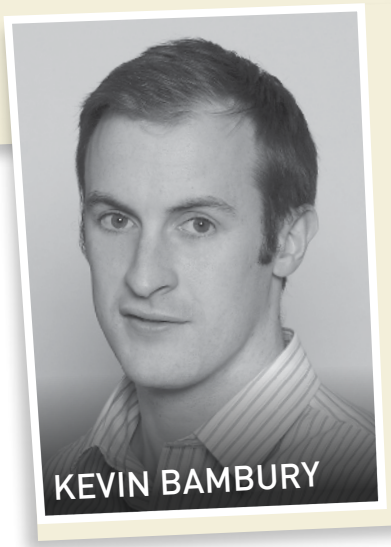
The word 'software' was specifically added to the MDD in 2007 and many apps qualify as devices. In response to continuing ambiguity, in January 2012, MEDDEV2.1/6 'Guidelines on the Qualification and Classification of Stand Alone Software used in Healthcare within the Regulatory Framework of Medical Devices' was published.

Also in January 2012 the Mersey Burns app, a clinical tool for estimating burn area percentages, prescribing fluids using Parkland, background fluids and recording patients' details, became the first MHRA approved Class I medical device app, thus paving the way for other developers to follow. But the question was, how to gain this approval?



Rhona is a microbiologist with expertise in GMP, medical devices and early-phase clinical research and has hosted MHRA, FDA and National Health Surveillance Agency (Anvisa) regulatory inspections.

With over 15 years' industry experience in QC/QA, Rhona is now a Principal Associate for TMQA, an independent quality assurance organisation providing worldwide audit, consultancy and training support.



Kevin is Co-founder/Director at Portable Medical Technology. Kevin graduated from the Masters of Electronic Business at the University College Cork with a first class honours degree. He has previous experience working in start-ups as the co-founder of Dingadeal.com and having worked with Sigojoven.com in Spain. He worked with AIB in brand management and marketing focusing on the area of sponsorships. He managed AIB's sponsorship of the Gaelic Athletic Association (GAA) club championships in 2007. Prior to this he studied a BA in English and Geography and a Higher Diploma in Management and Marketing.

ONCO ASSIST™
DECISION SUPPORT SYSTEM

One developer (Portable Medical Technology) developed the OncoAssist app - the application contains all the prognostic tools and useful calculators that oncologists need at the point-of-care. From a non-medical device background, they strongly believed that software used in hospitals to make clinical decisions must be developed and maintained in a controlled environment and pass the rigours of CE approval, the same as any traditional medical device such as scalpels or pacemakers. But to get to market, they had to navigate the unfamiliar waters of the regulatory environment, finally implementing the EC declaration of conformity route as described in Annex VII of the MDD. In that, they as the manufacturer, satisfied themselves that the product was a Class I device and it complied with the relevant essential requirements of the directive. They then drew up a written statement to this effect and registered with the competent authority (in this case the Irish Medicines Board (IMB)). The CE mark was then applied to the product and the product placed on the market.

As Kevin Bambury, Director at Portable Medical Technology explains – "the process at first was daunting, being totally unfamiliar with the regulations and the regulators, it was hard to know where to begin. Just getting to talk to the IMB about device classification was a minefield. However, once we got into the process, even the amount of documentation required began to make sense and it wasn't so painful. And now we are the first Irish and only the third company worldwide to be granted a CE mark for a smartphone app. The application of the CE mark has got to be a step up from the mentality of anyone publishing an app that can claim to do anything without concrete evidence. Doctors can't be expected to spend hours sifting through the appstores looking for the most reliable tool, with nothing to guide them but a star rating and some screenshots".

The days of untrustworthy, unapproved and unprofessional medical apps are coming to an end; the industry has to begin embracing the new age of regulation resulting in approved apps. Now comes the fun part, educating consumers as to what the CE mark actually means!

There are stringent compliance requirements for apps in terms of EU standards and FDA regulations relating to medical devices. Annex 1 of the Medical Devices Directive 93/42/EEC, medical devices must not only be safe but also function in a medical-technical way as described in the manufacturers `intended purpose'. Compliance with these requirements must be evidenced within a certified quality management system according to ENO ISO 13485. Information on applicable EU standards can be found on the European Commission website - **http://ec.europa.eu/health/medical-devices/documents/guidelines/index_en.htm** and FDA regulations on the FDA website - **http://www.fda.gov/medicaldevices/productsandmedicalprocedures/ucm255978.htm**



Data Display Considerations for Medical Device Data Systems

(FDA – Code of Federal Regulation Title 21 Part 880)

FRASER SMITH



The United States of America Federal Register, Volume 76, Number 31 dated 15th February 2011 published a new final rule for an electronic medical device known as Medical Device Data System (MDDS). An interpretation of this final rule is the subject of this article. As normally done with new regulations, the FDA prepared a written preamble published in the Federal Register concurrently with the final rule.

The preamble provides narrative discussion and the FDA responses to public comments received during historical development of the regulation. Preambles appearing in the Federal Register can provide valuable insight or clarification about the FDA's thinking with regard to various regulatory matters. In this way, preamble commentary can provide important information which helps to guide development of regulatory strategy. However, caution must be used when preamble content is considered, because it does not represent official FDA regulation nor does it represent enforceable requirements of the FDA. Nonetheless, preambles appearing in the Federal Register frequently influence the design of regulatory compliance programmes because of the importance of the information.

MDDS regulations are defined in 21 CFR 880 and evolved over the extended timeframe of 2007 to 2011. Any regulation developed over such a protracted interval represents a significant challenge for a regulatory agency. Over the timeframe of 2007 to 2011, numerous technological advances occurred in computer systems and networks. As a result, the technical context under which the earlier MDDS regulation was developed had to be re-evaluated to consider these changes.

Computer technology is characterised by rapid change and has frequently been described by a rule-of-thumb known as Moore's Law. The co-founder of Intel Corporation, Gordon Moore, made a whimsical prediction in 1965 that the number of transistors on a chip should be expected to double approximately every two years. An obvious impact of Moore's Law was the rapid, ongoing development of microprocessor and digital circuit technology. Figure 1 provides additional information about Moore's Law and the historical growth of transistor count in computer chips.

Intel has successfully maintained a rapid pace of development for almost 50 years under the predictions made by Moore's Law. This rapid technology development is unprecedented and has served as the overall springboard of Intel's business pursuits related to computer chips. The speed and capacity of computer chips is directly influenced by the density of transistors on chips. The rapid increase in speed and capacity of computer equipment these past 35 years is a result of Moore's Law.

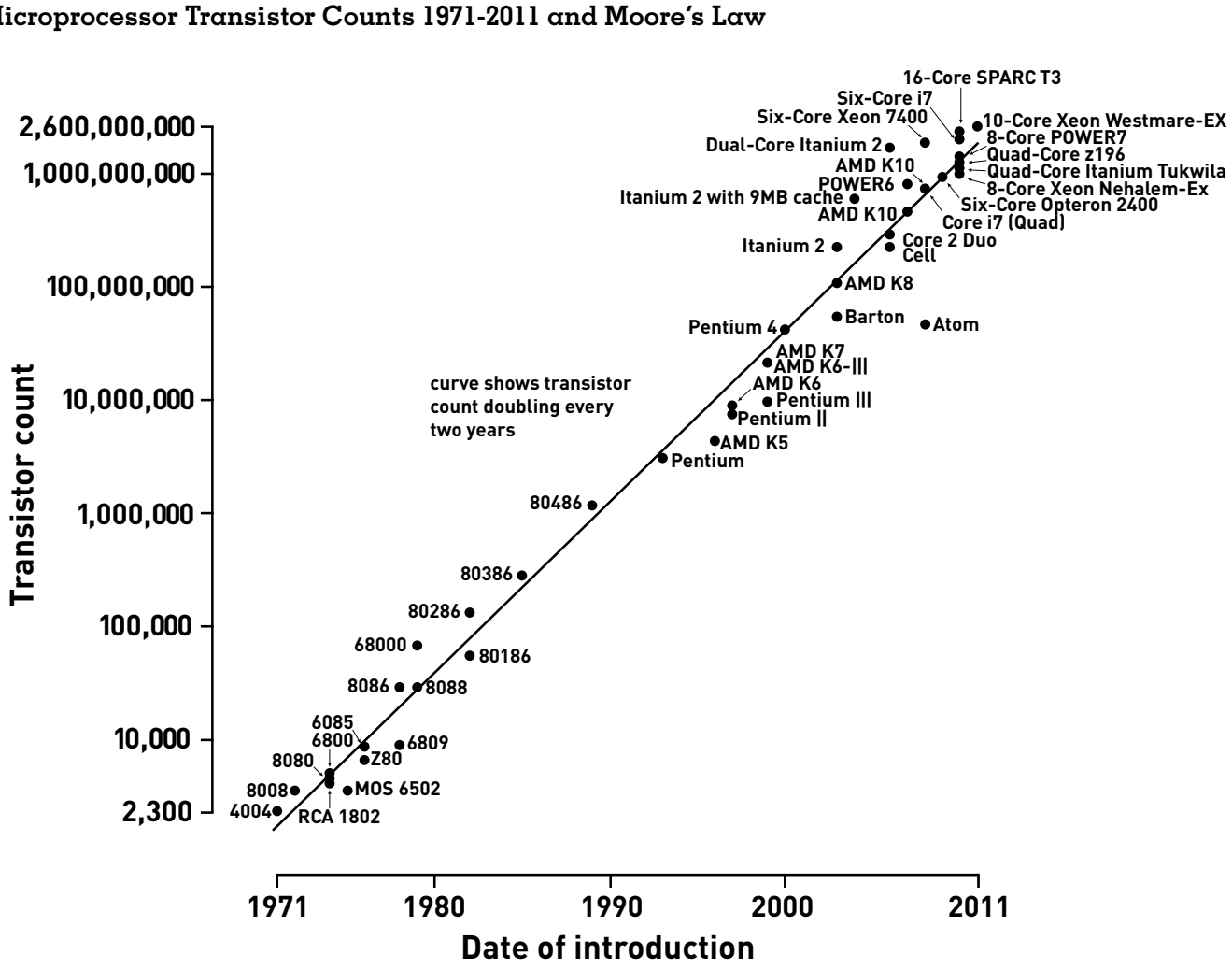
Under conditions of such rapid change, the FDA was obligated to develop a flexible yet adequate regulation for an MDDS which considers a wide spectrum of technology. Medical devices using computer technology may be a few years old, yet includes newer technology which has just entered the market. As such, development of a specific regulation for MDDSs has been a difficult challenge for the FDA. The regulatory approach of the latest update to the MDDS regulation has been to focus more on functional requirements of these devices rather than a specific technology requirement.

The Case of 'Real Time' Data Processing

In an earlier version of the regulation proposed for MDDS, the FDA considered 'real time data processing' to be an important safety consideration. However, as a result of Moore's Law, microprocessors became faster and more capable, and virtually all computerised equipment performs 'real time data processing'. Thus, an earlier concern about this technical characteristic was no longer a safety concern.

The FDA re-evaluated the safety concern represented by the technical requirement of 'real time data processing'. This re-evaluation apparently concluded that 'real time data processing' is no longer a safety concern. References to real time data processing are removed from the final rule.

Figure 1 – Historical Timeline of Transistor Count in Relation to Moore's Law



However, the FDA instead realised the functional requirement to prohibit 'active patient monitoring' is appropriate (see comment 19, p 8644 of FR Vol. 76, No 31 dated 15th February, 2011). A complete explanation of this term is beyond the scope of this article, but the important point is that many specific technical characteristics of MDDS have been replaced by functional performance characteristics.

In situations where a computerised medical device provides 'active patient monitoring', it represents a primary indicator of patient status that health care professionals will use for making treatment decisions. If equipment performs 'active patient monitoring', the FDA states the equipment cannot be considered MDDS which is exempt from premarket notification (Comment 19, p 8644 of FR Vol. 76, No 31 dated 15th February, 2011). This is the first example of a modification made to the MDDS regulation to consider improvements in technology.

The Case of Irreversible Data Compression

A second example pertains to the FDA's earlier concerns about irreversible data compression. A few years ago, microprocessors were slower and had less processing capacity. Under these technical constraints, software algorithms that compress digital data could also add a significant amount of distortion to medical signals. It was believed that this distortion could adversely affect the quality of diagnostic data used to make medical treatment decisions for patients and represent a safety concern.
The FDA considered a specific technical requirement in earlier versions of the MDDS regulation to restrict or prohibit any form of data compression which caused an irreversible loss of data. For devices which require data compression that result in irreversible loss of data, the FDA previously stated that these devices could not be considered an MDDS and instead requires review under a 510(k) premarket notification. This is significant, because the FDA allows an MDDS to be exempt from 510(k) premarket review.

As a result of Moore's Law, dramatic improvements in the speed and capability of computer chips have virtually eliminated earlier concerns about irreversible data compression. Furthermore, irreversible data compression is not likely to have any effect on medical treatments received by patients because the FDA has determined that an MDDS must not be used for 'active patient monitoring' as described previously. Therefore, the FDA eliminated the earlier technical consideration about irreversible data compression for MDDSs (Comment 17, p 8644 of FR Vol. 76, No 31 dated 15th February, 2011).

The Case of Numerical Data Display vs. Graphical Data Display

Two previous examples describe specific technical requirements earlier proposed for MDDS that were revised to reflect a more appropriate functional requirement. These examples show that the FDA has needed to be flexible and accommodating in development of the final rule for MDDS. It is expected that the FDA will need ongoing flexibility in their interpretation of the MDDS regulation. In this context, we now take a closer look at the primary consideration of this white paper. In preamble reply to Comment 10 (p 8642 of FR Vol. 76, No 31 dated February 15, 2011), the FDA discusses a specific technical consideration of MDDS about the form of data display.

The form of data display is considered a specific technical requirement which would pertain to MDDS. As mentioned above, the FDA removed specific technical requirements from the final rule. In reply to Comment 10 on p 8642 of Federal Register Volume 76, No. 31 dated 15th February, 2011, the FDA addresses a comment made to an earlier version of the MDDS regulation released several years ago. The content of the MDDS regulation was changed considerably since its last revision. The context of this information is somewhat confusing in the preamble as well as applicability to the final rule. The final rule does not include any regulatory requirements about forms of data display, so interpretation is necessary.

The FDA describes how an MDDS may convert or handle data in the preamble, which is a rather specific technical requirement. For situations where MDDS provides data conversion to allow communications in different languages, the FDA explains that data conversion which improves interoperability of equipment from different vendors is permissible. However, this is not described anywhere in the final rule.

An example of this is where one instrument provides data messaging via Health Level 7 (HL-7), the international standard on interoperability of health information technology, while another provides data messaging as comma separated values (CSV or 'spreadsheet data'). The specific technical example selected is less important than the FDA's rationale about why it's acceptable. Technical features which improve performance or provide interoperability are considered acceptable for MDDS. From a software engineering perspective, language conversions described by the FDA in the preamble are 'more complicated' (in terms of the amount and the type of information processed) than a conversion of numerical data to graphical display. Therefore, the complexity of implementing a technical requirement does not seem to be the basis for the FDA's concern about the function of an MDDS.

The FDA allows a MDDS to translate data messaging information to achieve better interoperability of hospital monitoring equipment. However, the FDA expresses concern about the format of data display to users. Like language translation, this is a specific technical requirement. This data display consideration is considerably 'less complicated' to implement from a software programming perspective than language translations mentioned in the earlier example.
As above, this further substantiates that the complexity of implementing a technical requirement is not the basis for the FDA's concern about the function of an MDDS.

It's somewhat confusing that the FDA would permit a more complicated and potentially more risky language translation to be allowed for an MDDS, yet prohibit a less complicated and less risky graphical display of numerical data for an MDDS. This is perhaps clarified by understanding that the preamble provides explanations related to prior comments made with regard to earlier revisions of the MDDS regulation, and does not specifically provide explanations that apply to the final rule. It is also helpful to understand these are merely examples that were discussed to explain why certain requirements were implemented.

The FDA's concern about data display apparently relates to the agency's general concern about computer screen interfaces. This wasn't explained in the preamble. The FDA allows improvements which increase usability (for example language translations) but discourages changes which may make equipment more difficult to use. Looking beyond the preamble at well-established requirements for graphic user interfaces that the FDA has issued for human factors/usability, clarification is found about expectations for data displays.

Human factors/usability considerations are described in the FDA Guidance Document No. 1757 that is used as benchmark evaluation for all forms of graphical user interfaces. This guidance document has been used to evaluate hundreds of graphical user interface designs and has become recognised as the 'gold standard' of the FDA. It's rather certain that the FDA has no intention of implementing a new guideline or requirement for graphical user interfaces in the preamble of the MDDS regulation.

The FDA relies on human factors testing to assure that computer screen displays are easy to understand and unlikely to result in misinterpretation of data. The FDA does not address human factors testing in final rule of 21 CFR 880. When the FDA makes a preamble comment about displaying numerical data in graphical form, it is implied that they are considering this from the human factors/usability perspective.

From prior experiences with the FDA regulation, it seems apparent that the FDA is concerned that user perception of numerical data may be different than the perception of graphical data. In instances where a device sponsor has performed a human factors/usability test, however, test results will show that users can properly decode data display information (whether it's in numerical or graphical form).

Understanding that the FDA has moved away from describing specific technical requirements for MDDS in the final rule (largely a result of Moore's Law which makes development of specific technical requirements difficult), confusion resulted when the FDA explains specific technical requirements in its reply to Comment 10. Nonetheless, if a device sponsor can show that users can properly interpret medical data on screen displays, whether or not presented in numerical format or graphical format, then there's no reason for the FDA to prefer one format over the other. The FDA has issued a formal guidance document that addresses this matter in considerable detail.

Therefore, a sponsor who performs human factors/usability testing of its screen displays during validation will be able to demonstrate that numerical or graphic displays are equivalent. On this basis, the FDA would not deny MDDS status of a particular computerised device when substantial equivalence of screen display interpretation has been performed in a human factors evaluation. This is no different than the expectation that would apply to a 510(k) if one were prepared.

Conclusion and Recommendation

In this discussion, the background and regulatory significance of preambles of regulations are discussed. Preambles provide insight into the FDA's thinking and rationale for a particular regulation, but do not provide a specific regulation subject to enforcement. Requirements that device sponsors must follow are defined in regulations rather than preambles of regulations.

The FDA has provided a preamble with the MDDS regulation. An example of numerical versus graphical display of data is provided in the preamble with respect to permissible functions of an MDDS. However, there's no specific requirement pertaining to data display in the actual MDDS regulation. The FDA normally evaluates the suitability of graphical displays through human factors/usability testing.

From the discussion provided in the preamble, the FDA finds improvements in MDDS performance (such as language translations) which allow better interoperability are acceptable. Likewise, if a sponsor shows that display of numerical data in graphic formats is properly understood by users, it would likewise be acceptable.

The FDA's preamble discussion point about graphical displays apparently relates to a human factors/usability consideration. Further clarification in this regard is found in the FDA's Guidance Document No. 1757 which pertains to design of user interfaces. Guidelines for performing human factors/usability testing are described in the FDA's guidance document.

Since an MDDS is a type of computerised medical device that would otherwise be reviewed as a premarket notification 510(k), it's appropriate that a device sponsor consider how these submissions are prepared. In a typical 510(k), a device sponsor presents data and other information to show that the device under review is substantially equivalent to a predicate device. These 510(k) presentations show the functional equivalence of devices by comparison of performance features and intended use. The FDA's human factors/usability guidance document has been used extensively to evaluate computerised medical devices submitted under 510(k)s and assuredly represents a suitable method to evaluate a medical device screen display/user interface.

If validation testing of MDDS includes an evaluation of the screen display/user interface under the FDA's guidance document for human factors/usability and the presentation of medical device numerical data is shown to be substantially equivalent to the presentation of graphical data, then every reasonable expectation has been fulfilled.

The final rule for MDDS (21 CFR 880) does not contain any regulations that prohibit presentation of numerical data in a graphical format, or the presentation of graphical data in a numerical format. It's presumed that the device sponsor has implemented a quality system which complies with 21 CFR 820, in particular the requirements pertaining to design control.

It's expected that validation testing of MDDS is performed under a quality system which is compliant with the FDA's quality system regulation, a requirement for MDDS. If validation testing demonstrates that requirements of the MDDS regulation and device specification are met, then the device sponsor will satisfy their regulatory obligation.

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Moore's Law http://www.mooreslaw.org/

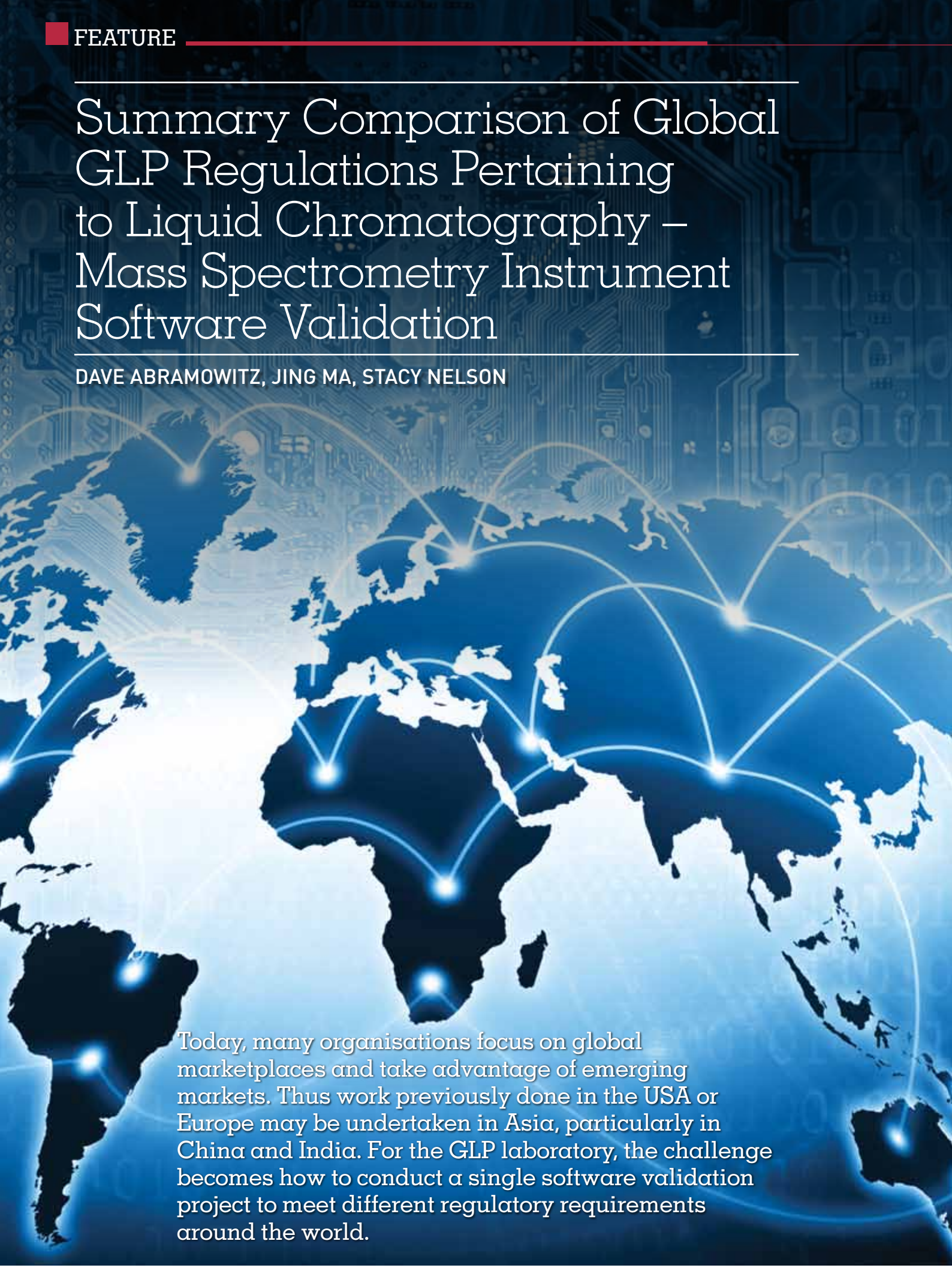


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Summary Comparison of Global GLP Regulations Pertaining to Liquid Chromatography – Mass Spectrometry Instrument Software Validation

DAVE ABRAMOWITZ, JING MA, STACY NELSON



Today, many organisations focus on global marketplaces and take advantage of emerging markets. Thus work previously done in the USA or Europe may be undertaken in Asia, particularly in China and India. For the GLP laboratory, the challenge becomes how to conduct a single software validation project to meet different regulatory requirements around the world.

Figure 1: Applicable Regulations Coupled with Risk Assessment Allows a Global Validation Approach



The key is interpretation of the applicable GxP regulations and guidance coupled with effective risk management. This article summarises the AB Sciex whitepaper titled, *Comparison of Global GLP Regulations Pertaining to Liquid Chromatography – Mass Spectrometry (LC/MS) Instrument Software Validation.* It considers the similarities and differences between the following regulations:

- United States Food and Drug Administration (FDA) and GLP regulations under 21 CFR 58 and 21 CFR 11
- Japanese GLP issued under Ministry of Heath Labour and Welfare (MHLW) Ordinance 21
- Organisation of Economic Cooperation and Development (OECD) GLP regulations adopted in the majority of OECD member countries
- World Health Organisation (WHO) GLP Handbook principles.

Figure 2 shows the relationships between these regulations and other guidelines issued by the FDA, OECD and Swiss AGIT (Working Party for Information Technology).

United States Food and Drug Administration (FDA) GLP

Under GLP Regulations for Non-Clinical Studies, 21 CFR 58[1] software is considered part of equipment designed for specific use. Thus, there are three main requirements:

- Appropriate design
- Suitable location
- Adequate capacity.

Additionally, for computerised system equipment, Electronic Signature and Electronic Records, 21 CFR 11[2] requires three types of control: administrative, procedural and technical such as the security and integrity of system generated data and electronic signatures.

The FDA also issued industry guidance including General Principles of Software Validation[3]. The principles addressed here include:

- Planning is essential for computerised system validation
- User requirements are mandatory to define the intended purpose of the system
- Requirements must be traceable throughout the life cycle
- Testing needs objective evidence documented to support claims that the software meets user requirements.

Japanese GLP Regulations

The Japanese GLP regulations were issued under Ordinance 21 by the Ministry of Health, Labour and Welfare[4]. Similar to the US GLP regulations, there is the requirement for equipment to be designed appropriately with sufficient processing capabilities.

OECD GLP Regulations

OECD is responsible for GLP regulations used in member counties (except the USA and Japan) for non-clinical testing studies[5]. Like other GLP regulations the apparatus under OECD must be suitably located, designed appropriately and have adequate capacity. The OECD has also produced industry guidelines[6].

Additionally, the Swiss AGIT is a collaboration of Swiss inspectors and pharmaceutical industry experts. They have published four practical guidelines since 2000 [7-11].

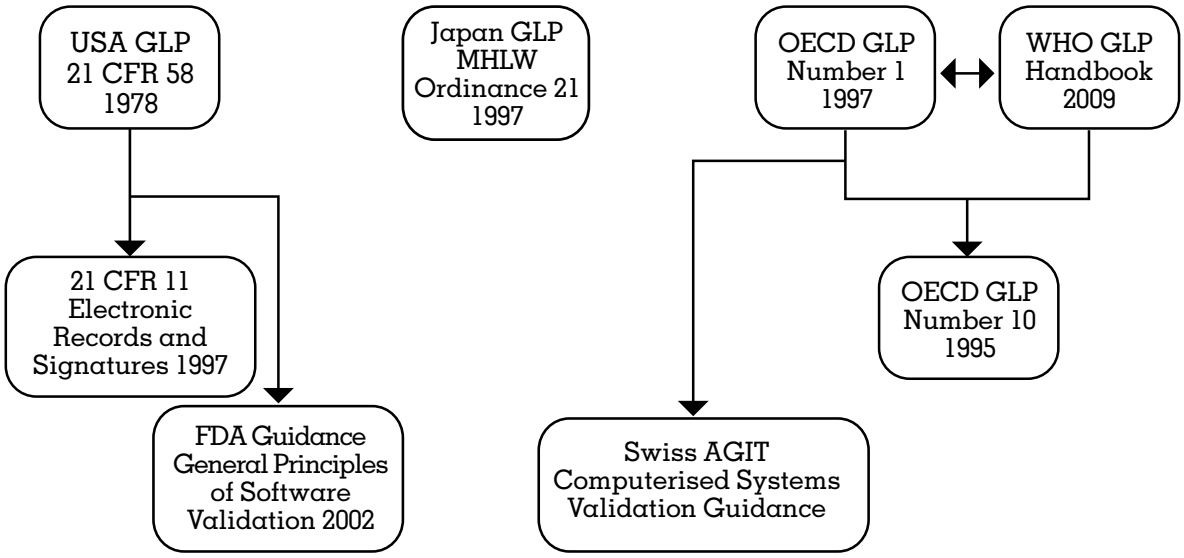


Figure 2: Regulatory Guidance

WHO GLP Handbook (2009)

The WHO GLP Handbook[12] interprets GLP regulations from the OECD. The section 'Facilities, Buildings and Equipment' addresses equipment including computerised systems.

Industry Guidance

Early guidance on the use of computerised systems in GLP environments was the 1988 publication Computerised Systems Used in Non-Clinical Safety Assessment (Red Apple Document)[13]. This was updated in 2008[14] but it disregards risk management, user requirements and traceability of user requirements throughout the life cycle. Therefore, it missed the point of life cycle validation.

Modern Good Automated Manufacturing Practice (GAMP® 5) Risk-Based Validation Guidance addresses risk management[15]. This is important because the GLP laboratory may only use 20% of the functions of the software. So, why validate those not used?

Global Validation Approach

In conclusion, to ensure a validation project meets global GLP regulations, consider the factors listed below and the life cycle in Figure 3.

- Risk management
- Planning and reporting the validation effort
- Specifying user requirements and system configuration
- Qualifying supplier
- Ensuring traceability of requirements throughout the life cycle
- Installing, integrating and qualifying the components to form a system
- Testing the configured system meets user requirements
- Writing procedures and training users
- Releasing the system for operational use.

System Validation Documents

The following documents would meet global GLP requirements:

- Validation risk assessment (VRA) – risk assessment based on GAMP® 5
- Validation plan (VP) – comprehensive plan for conducting validation that defines scope, roles and responsibilities, life cycle and documented evidence
- User functional requirements specification (UFRS) – list of user and functional requirements based on the GAMP® 5 risk-assessment
- System design and configuration specification (SDCS) – technical description of the system
- Test plan (TP) – list of tests to be conducted
- Design qualification (DQ) protocol – verification that software was designed to meet intended use
- Installation qualification (IQ) protocol – test scripts to verify that software has been installed and configured as per design specifications
- Operational qualification (OQ) protocol – test scripts to verify that software operates in accordance with design specifications
- Performance qualification (PQ) protocol – test scripts to verify that software consistently meets performance specifications when operating under normal conditions
- Traceability matrix (TM) – method of tracking the user and functional requirements of the test scripts
- Validation summary report (VSR) – comprehensive summary of the results of validation activities.

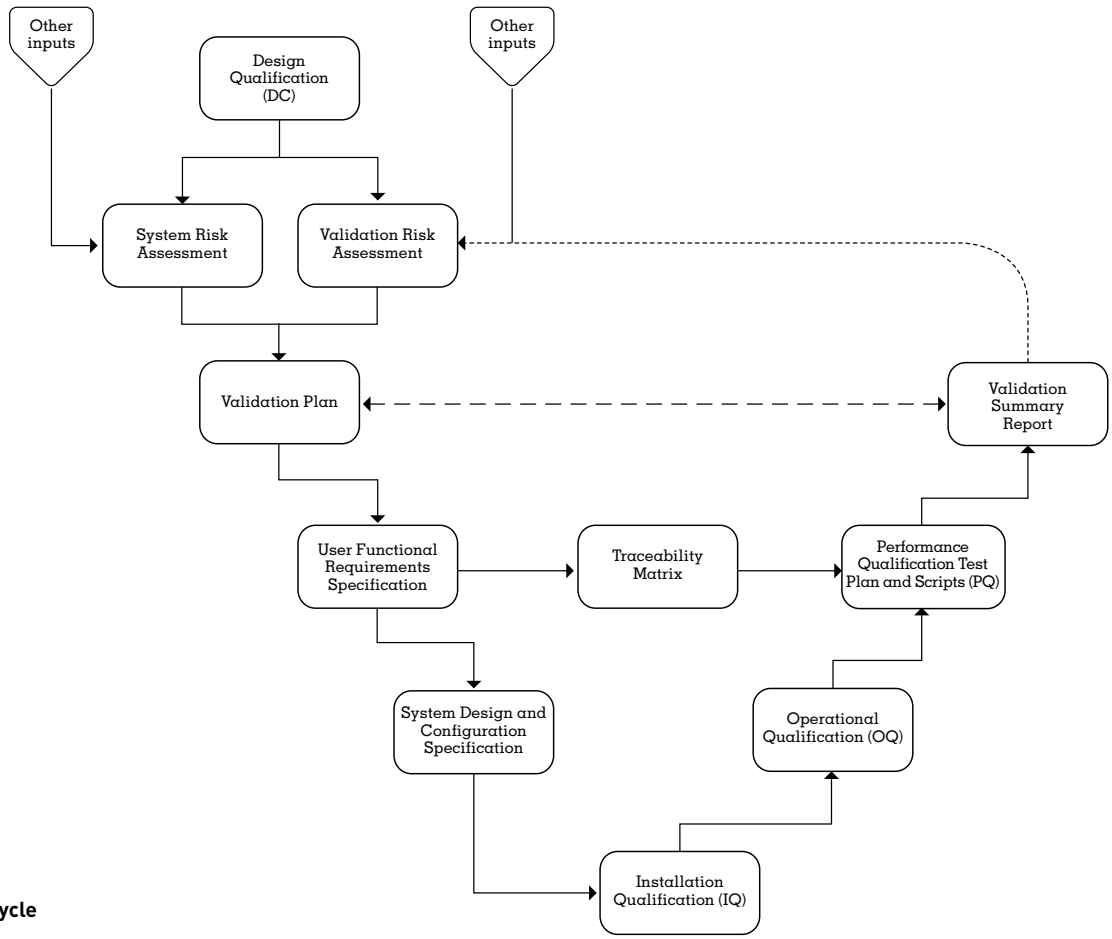


Figure 3: System Life Cycle



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As the Senior Manager for the AB SCIEX Compliance Services and Validation Consulting organisation, Dave has led the group since its inception in 2006. He currently travels the globe to work with AB SCIEX's mass spectrometry users in mapping out and designing compliance strategies. For nine years prior, Dave was with Cisco Systems in various IT and operations roles, where he achieved his PMP® certification. Dave also has extensive experience in the financial derivatives markets working for various banks and trading firms in New York and Chicago. Dave has a BA in Economics and an MBA from the University of Texas in Austin.



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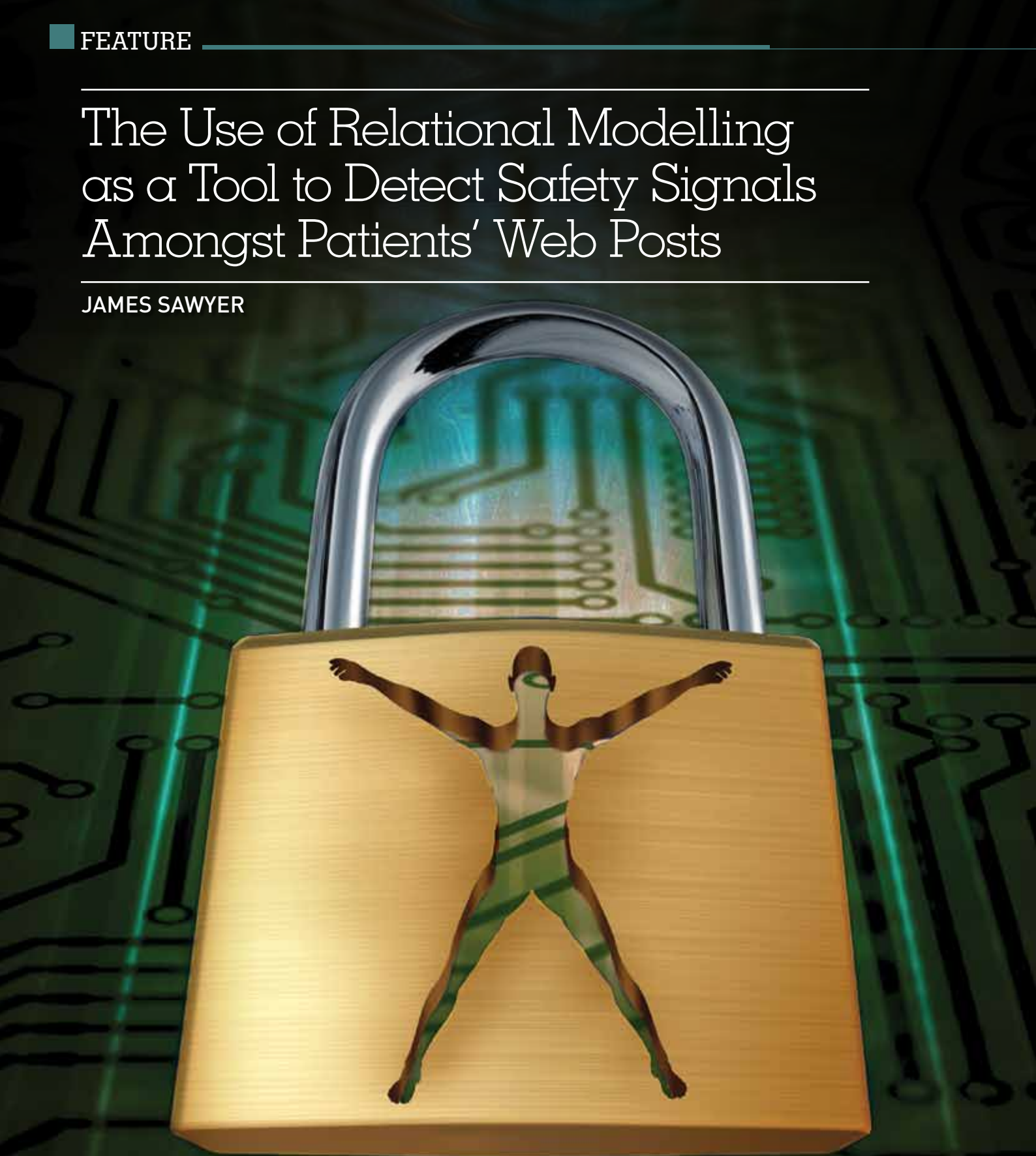
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The Use of Relational Modelling as a Tool to Detect Safety Signals Amongst Patients' Web Posts

JAMES SAWYER



Traditional methods for collating product-related safety information are effective for structured data collected through clinical research and also for semi-structured data captured through national spontaneous reporting processes.



The published literature is also a manageable source of safety information for signal detection and can be mined using electronic repositories and search engines such as PubMed and Embase. However, the healthcare industry has an obligation to evaluate all data sources; those within its control and potentially those beyond.

The majority of this additional information is present in the form of free text, either within internal company reports or spontaneously posted on the internet. It has become impossible to manually monitor the huge volume of information that is continuously generated. Automation of this task is the only remaining viable option for regulators and pharma alike if they are to capitalise upon the public health opportunities these 'big data' present.

Signal Detection within Unstructured Data

There are three steps involved in the analysis of unstructured data for the purpose of assessing health outcomes:

- Non-specific data capture and aggregation
- Data sorting and enrichment
- Contextual analysis and outcomes reporting.

For the purposes of pharmacovigilance 'no stone should be left unturned'; hence, each individual data source reviewed for signal detection should be as comprehensive as possible. Where this relates to internal company documents, there is a natural limit to the data; the volume of information is finite and usually the data items will be independent (e.g. reports of individual personal interactions). With internet-derived data there are different issues, since the volume of potential sources is almost infinite and there are challenges to understanding its provenance. Each of these challenges needs to be overcome in order to create a database that is a valid and representative repository of reported experience.

Data sorting and enrichment are the linchpins of effective semantic analysis. Simplistic approaches have been highly effective in many areas, for example in terms of hotel and restaurant rating. In these situations, a mixture of sentiment analysis and keyword capture provides the basis for global assessments of user experience which, when collated, yield a semi-quantitative measure of quality. When considering patient-treatment experience and outcomes, this approach is too blunt to yield specific and relevant information. The natural language analysis involved must, at a minimum, identify the potential for a cause-effect relationship between product use and outcome. In its most basic construct, this is about exposure to product preceding the reported experience, but also requires that extraneous unrelated information is censored. Relational modelling achieves this goal and helps to improve the specificity of the data-mining process and remove 'noise' from the system. Conversely, the sensitivity of the system to detect relevant health-outcomes information is increased beyond that afforded by simple keyword identification by using broader language to detect outcomes such as adverse events (AEs).

Mapping this natural language to conventional medical terminology such as the MedDRA dictionary of adverse-event terms is an essential step that allows information to be sorted into a format that is consistent with outputs from analyses of data from conventional structured sources. The final step in the data enrichment process is the integration of other information that is relevant to medical case evaluation, but which might not be contained within the individual 'snippet' from which an experience is captured. These data include age and gender of the reporter, aspects of their medical history and concurrent treatments and experience of other medications. The effectiveness of individual algorithms to detect and select such information depends upon the format of the data and use of language. Hence, to deliver the required sensitivity and specificity, different algorithms may be required to analyse reports made by company sales staff in templates compared with the data captured in patients' spontaneous web postings.

Current approaches to evaluate potential safety signals often involve direct interaction with the reporter. When working with passively captured or archive datasets, this is often not possible. Therefore the information about individual outcomes may vary in terms of 'richness', i.e. the amount of detail available compared with what would be desirable and indeed available in, for example, a clinical trial setting. Such incomplete case reports are subject to regulatory reporting once a company becomes aware of them. As a result, there is little incentive for either regulators or manufacturers to identify large volumes of almost inevitably incomplete reports that can only add to the level of noise and burden of work associated with their management and processing. Two processes must be applied to these data for it to have utility and value for signal detection purposes. First, as for an observational study, individual reports must be collated so that data can be summarised. Second, the data counts need to be placed into the context of overall AE reporting numbers and the condition the product is being used to treat; the concepts of relative and proportional reporting ratios are widely accepted in the field of drug safety assessment. With access to large, clean datasets and appropriate tools to identify AEs possibly associated with treatment, these ratios can be estimated and potential safety signals identified. Incorporating the broader panel of information that is available within internet posts means that confounding factors can be considered. As a result, this technique may be used for more than simple hypothesis generation.

Proof-of-concept studies have indicated that this novel approach has the capacity to detect outcomes that have a high impact on patients in clinical practice, but did not feature as major clinical concerns within clinical development programmes. For example, using this approach, the occurrence of neuropsychiatric AEs in association with varenicline among patients ceasing smoking was easily identifiable in web postings made in the first few months after launch. Similarly, for more mature products, the sheer volume of patient experience reported on the internet facilitates identification of rare events or events that occur at an increased frequency in particular clinical circumstances.

Next Steps

Progress in this area requires a tripartite collaboration between the drivers of innovation, the pharmaceutical companies which have a responsibility to evaluate the safety of their products in clinical practice and the regulatory agencies responsible for public health. For the innovators, it is essential that they develop robust methodologies that yield meaningful and reliable information compatible with existing methodologies. Pharmacovigilance teams need to embrace the opportunities afforded by new technology and to help characterise their place within the armamentarium of signal detection tools available. For the part of the regulators, they have a critical role in nurturing the use of new approaches and providing a framework that facilitates the early use of novel data sources.

'for more mature products, the sheer volume of patient experience reported on the internet facilitates identification of rare events or events that occur at an increased frequency in particular clinical circumstances.'

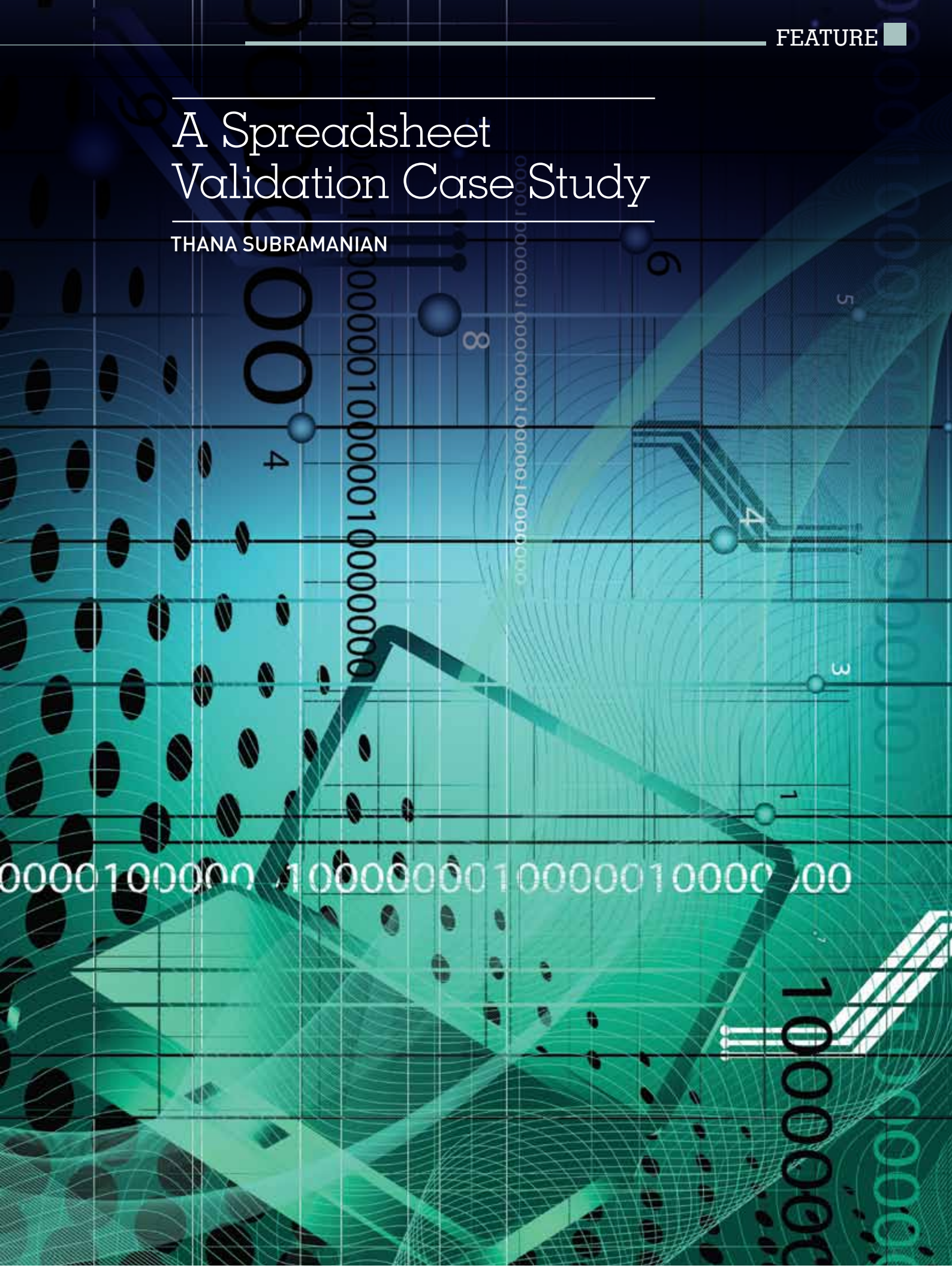


JAMES SAWYER

James joined the pharmaceutical industry 20 years ago as a Clinical Research Physician. He alternated between medical and commercial roles at big pharma companies such as Sanofi, AstraZeneca and Roche before establishing Prism Ideas in 2001. He has a long history of innovation, ranging from clinical trial feasibility through research into stratified care and approaches to clinical trial design. More recently, he has been exploring how the internet can be used as a resource for pharmacovigilance.

A Spreadsheet Validation Case Study

THANA SUBRAMANIAN





Regulatory/Guidance Inputs

- 'Spreadsheet Design and Validation for the Multi-User Application for Chemistry Laboratory' Part I, No. 4317, Laboratory Information Bulletin Division of Field Science, Office of Regulatory Affairs, FDA
- 'Spreadsheet Design, Verification and Validation, Use and Storage of Single-User Workbook Files in the US FDA Laboratories' Part II, No. 4349, Laboratory Information Bulletin Division of Field Science, Office of Regulatory Affairs, FDA
- 'GAMP® 5 Risk Based Approach to Compliant GxP Computerised Systems'
- 'Guidelines for the Development and Validation of Spreadsheets', Working Group on Information Technology (AGIT), version 1, 1st August 2011

The Scenario

A small facility couldn't obtain approval for a laboratory information management system (LIMS) to be implemented, so the laboratory analyst decided to be creative and develop over 80 spreadsheets ranging from calculation to trending spreadsheets. Unfortunately, the analyst did not think about the need to validate these spreadsheets!

The Plan - How Did We Proceed?

We needed to know the true extent of the issue, so the first thing was to establish an inventory of the spreadsheets. Searching the servers and workstation hard drives was not going to be a solution as there was no way of discerning what was GxP or not. It was a case of interviewing departments within the facility. The laboratories were not the only creators of GxP spreadsheets, as we found facilities had a steam quality calculation spreadsheet too.

In parallel, a plan was being drafted to describe the remediation activities.

In addition, to ensure that other sites were learning from this discrepancy, a procedure and training material on spreadsheet validation were developed.

We proceeded to pilot two types of calculation and trending to see if there were any kinks that needed ironing out in the procedure and methodology.

We also had an underlying aim to simplify and minimise, if not eliminate, any customised code. Microsoft Excel has an excellent programming tool using a type of visual basic for applications (VBA), which can be created unwittingly via recording macro function.

Some Basics of Spreadsheets

A desktop application like Microsoft Excel is widely available on many users' workstations. From this application or tool a user can create a spreadsheet, which can be very simple or very complex. This spreadsheet is essentially a user created application. There are three types of spreadsheets that we can classify:

1. The single use, throw-away spreadsheet
2. The calculator spreadsheet
3. The data spreadsheet.

Control

Another complication was the control of both the data and calculation only spreadsheets.

The initial thoughts were to utilise our electronic document management system (eDMS) to control both types. However, the practicality of checking in and out a data spreadsheet, but also how to comply with the technical requirements of electronic records, in particular, audit trails without having to create any customised code had to be considered.

Two routes lay ahead of us. One took us down the development of customised code which could then be created either as a module that could be validated once and added to each spreadsheet or go down the route of purchasing a third party package and applying a GAMP® category 3/4 validation approach.

Rather than create custom code we took the decision and looked for an off-the-shelf solution, which provided 21 CFR Part 11, electronic records and electronic signature (ERES) technical solutions for audit trails and levels of security. This would provide a better audit trail, at a cell level, over an eDMS for data type spreadsheets. The wrapper solution also meant we didn't need to utilise an eDMS for the production environment and could use a secure server share. There are a number of solutions on the market that provide a 'wrap around' solution for the spreadsheet that will provide security and ERES technical compliance. The same technical solutions could be performed in-house by means of customisations, using Microsoft Excel's VBA programming tool. However this assumes the technical competence is available in-house.

Simplification

When reviewing and assessing all the spreadsheets, it became obvious that there were opportunities to combine some of these and reduce the validation activity.

After looking at the need to eliminate customisation, we were unable to eliminate all customisation; there was a need to create a small piece of VBA code for the trending spreadsheets.

In order to reduce the validation work we treated this as a 'module' that could then be validated once and integrated with all the data/trending spreadsheets.

The SOP/Validation Approach

The general approach to spreadsheet validation does not differ from the other validation approaches. The principles of specifications, planning, testing and reporting are the same. However, the scale and complexity of spreadsheets compared to larger business Information Technology systems allows us to reduce the validation activities to make it fit for purpose. This means the roles and deliverables are combined. For example, the owner is typically also the developer of a spreadsheet. The role of QA is important in the initial roll-out of this procedure, and it would be prudent to have greater QA oversight initially, but that can then be reduced for the more simpler and less risky spreadsheets.

The procedure provided detail on spreadsheet types and the approach to validating. It also had enough information to act as a validation plan for the simpler spreadsheets, with sections on:

- Risk assessment
- Requirements
- Design specification
- Configuration review or VBA code review
- Testing and reporting of unit, installation and functional/user acceptance testing.

Risk Assessment

The first level of risk assessment is to understand the GxP requirements for the spreadsheet and then to understand the system impact based on risk severity/probability/detectability.

Along with system impact, the validation approach can then be tailored to the spreadsheet depending on the GAMP® category of the component:

- Native spreadsheet function with no configuration = Category 3 (specify, installation and verify)
- Using calculations that reference other cells automatically to perform data manipulations = Category 4 (specify, design, installation, configuration review and verify)
- Custom macros or nested logic/lookup functions = Category 5 (specify, design, unit test, installation, code review and verify).

Requirements and Design Specifications

One of the concepts we employed was to use the spreadsheet tab functionality to provide the mechanism for documenting requirements and design. So a template spreadsheet was created which had a tab for specifications, a tab for installation, a tab for functional testing, a tab for VBA macros, a tab for the testing results and summary and then the final tab would be the actual application developed in the spreadsheet. This would mean that the details of the spreadsheet were kept and maintained with the spreadsheet.

Utilising the cells for requirements meant that design specifications were visibly traceable in the next column adjacent to the requirement statement and the next column can then identify the test.

	A	B	C	D	E
1	**USER REQUIREMENTS SPECIFICATION (URS) /DESIGN SPECIFICATION (DS)**				
2					
3	**User Requirement Reference**	**User Requirement**	**Design Specific Reference**	**Specification**	**Test Script Reference**
4	**URS 1**	**General Requirements**	**DS 1**	**General Design**	**IQ/OQ**
5	**The following are general requirements for the spreadsheet:**				
6	URS 1.1	All calculations can be viewed on one worksheet.	DS 1.1	The calculation part of the spreadsheet will consist of a single worksheet.	IQ 1.11
7	URS 1.2	The operator must be able to sign and date each page of the test results report in wet ink.	DS 1.2	There will be space at the bottom of each page clearly reserved for a operator's signature and date.	IQ 1.91
8	URS 1.3	The checker must be able to sign and date each page of the test results report in wet ink.	DS 1.3	There will be space at the bottom of each page clearly reserved for a checker's signature and date.	IQ 1.92
9	URS 1.4	The date after which the spreadsheet must not be used must be displayed in accordance with site procedure.	DS 1.4	The date after which the spreadsheet must not be used will be clearly displayed on the worksheet.	IQ 1.10
10	URS 1.5	The spreadsheet must be version controlled.	DS 1.5	The spreadsheet will be version controlled in the DeCO EDMS.	IQ 1.3
11	URS 1.6	Access to locked cells requires knowledge of the administrator password for the spreadsheet.	DS 1.6	Using Excel functionality to protect the worksheet, all users of the worksheet will be allowed to select and enter data into unlocked cells only. All other options under Tools>Protection>Protect Sheet shall be deselected. The administrator's password is required to unprotect the worksheet.	IQ 1.33 IQ 1.34

Figure 1: General Requirement Examples for a Spreadsheet



Some of the general requirement example for a spreadsheet are identified in Figure 1. Some of the best practices for spreadsheets are available in spreadsheet validation literature, for example:

- Locking cells other than areas of data input; calculation cells must not be editable by a normal user
- The use of colour coding for different parts of the spreadsheet, i.e. data input cells are a different colour to cells displaying results
- Using a standard layout for your spreadsheet will facilitate its maintenance. Including tabs, as described previously for the documentation, a menu, data input, calculations, results
- Provide clear units for data inputs and where rounding is required, ensure these are clearly defined in the requirements.

Configuration and Code Reviews

To ensure that the macros and/or formulas are correct, verification is performed against the specification. Where VBA is created for complex design specifications, the code reviews must be performed by an independent technical expert. Ensure that the formulas and configurations that are captured for review are retained as part of the validation documentation.

Testing and Reporting

Test scripts need to cover both installation and functional/acceptance testing. Installation testing should cover evidence in a controlled secure environment, where the 'wrapper' was used, the setup of security parameters within the secure network area must recorded. The functional testing must test all functions defined in the user requirements specification and design specifications. The amount/level of testing will be based on the categorisation and system impact.

Before completing the report to release the spreadsheet into production, a number of procedures need to be in place, apart from the obvious usage standard operating procedure (SOP), administrations, including access control, error logging and change control SOPs need to be in place. The majority of these procedures can cover multiple spreadsheets.

Conclusion

There is a lot of guidance available when it comes to spreadsheet validation. Developing an internal approach and piloting it first will help refine the approach, for security and electronic records and signature compliance, decide on customisation or purchasing a wrapper package to provide compliance.

Some of the ongoing challenges that need to be considered are continued ownership, maintenance activities, such as change control and archiving. Under change control, it is important to address new versions of Microsoft Excel and the impact on the spreadsheets. Archiving could pose a challenge for changing versions of Microsoft Excel if the data is kept electronically rather than printed off.

Microsoft Excel and other spreadsheet tools are readily available to the end user and these can become GxP applications, without the usual controls of validation. A good education programme and a simple to use spreadsheet validation procedure and templates are key in achieving compliance.



THANA SUBRAMANIAN

As QA Director, Software and Instrumentation, Thana has a global responsibility as system owner for eDMS and TrackWise (change control, IR/CAPA, complaints) systems. In addition, Thana provides project support, consultancy and auditing of GxP computerised systems for the medical diagnostics, GE Healthcare.

Thana has been a member of the RQA Computing Committee since 2006 and is also a ISPE UK GAMP® Community of Practice Steering Committee member and GAMP® R&D Clinical Systems Special Interest Group member.

Veterinary Quality Assurance and Computer Systems Validation

IAIN MCPHEE



GLP has been a requirement for veterinary target animal safety studies as long as computerised systems have been in widespread use within the laboratory. Sophisticated computerised tools have replaced the presentation of data using slide rule, graph paper and coloured pens in the space of a working career. Computers and their programs have steadily developed and will continue to do so into the future.

But still computer systems validation (CSV) and computer systems compliance elicit fear and foreboding in the hearts of many QA specialists. With the ever increasing use of technology, we need to be at ease and understand computer systems, their validation requirements and our role as QA in the process.
Increasing use of computerised systems in veterinary R&D is seen in the adoption of:

- Electronic document management systems
- Electronic data capture systems
- Data management systems
- Laboratory information management systems
- Electronic QA audit management tools.

Interfaced computerised systems automatically transfer information from one system to another in supra integrated networks providing additional challenges in validation. QA in veterinary R&D needs to rise to the challenge of a changing technological environment.

Regulations and Guidance

Veterinary GCP, veterinary pharmacovigilance and GLP require computer systems to be fit for purpose and therefore validated. Without validation the quality of computer generated data cannot be assured.

GLP Monograph 116 says, *'Where these computerised systems are associated with the conduct of studies intended for regulatory purposes, it is essential that they are developed, validated, operated and maintained in accordance with the OECD Principles of Good Laboratory Practice (GLP)'.*

VICH GCP less emphatically says, *'if data are entered directly into a computer system, the electronic record is considered the raw data. A computerised system should ensure that the methods for record keeping and retention afford* ***at least the same degree of confidence as that provided with paper systems****. For example, each entry, including any* ***change, should be made under the electronic signature of the individual*** *making the entry, and any changes that are made to data stored on electronic media should be* ***maintained in an audit trail*** *to protect the authenticity and integrity of the electronic records'*. Critically, VICH GL9 defines the now accepted requirement that if e-data is generated in computer systems the e-data is the raw data.

Veterinary Pharmacovigilance Volume 9B - Pharmacovigilance for Medicinal Products for Veterinary Use says *'d) Databases. A listing of the main databases used for pharmacovigilance purposes (e.g. compilation of safety reports, expedited/electronic reporting, signal detection, sharing and accessing global safety information) and brief functional descriptions of these should be provided including a statement regarding the validation status of the database systems'.*

FDA submission to USA requires GxP computer systems to comply with the Code of Federal Regulations (21 CRF Part 11), requiring attributes such as:

- Unique personal two part access, normally user name and password
- Periodic password replacement
- Lockout after repeated incorrect login attempts
- Re-login required after period of non use
- Audit trail as part of the meta data collected by the system to show who captured, changed or deleted data with a time stamp to show when they did.

Requirement

Where computerised systems are associated with the conduct of studies intended for regulatory purposes, it is essential that they are developed, validated, operated and maintained in accordance with regulatory requirements. In summary, each computerised system used in a GxP environment must be validated to assure confidence in the workings of the black box at the heart of the system.

What is Computer Systems Validation?

'Establishing documented evidence which provides a high degree of assurance that a computerised system consistently performs according to predetermined specifications and quality attributes'. FDA Glossary

'Validation of systems (should be performed) to ensure accuracy, reliability, consistent intended performance, and the ability to discern invalid or altered records' and *'Revision and change control procedures to maintain an audit trail that documents the time sequenced development and modification of the systems documentation'.* 21 CFR Part 11

'Nothing more than Good Software Engineering practices'. IEEE (ISO) 9001-3

Validation is scalable depending on the system risk, size and complexity.

Quality Assurance

Quality assurance however defined as QA, e-Compliance, Quality Engineer, etc., fills one of the five key roles for CSV:

- System owner
- Application manager
- Business owner
- IT quality manager
- Quality assurance – (titles and staff nos. involved depends on company and project size).

Compliance reviewing is the most significant aspect of the QA role, hence the requirement for independence from the IT function and segregation of duties between quality delivery (IT QM) and QA. QA involvement in CSV includes review and acceptance for the key validation deliverables.



Role of Quality Assurance in Relation to CSV

- Undertakes vendor audits, where required, to assure the process for software development
- Approves the high level risk assessment (HLRA) that defines if a computerised system is GxP relevant or not and therefore the need for validation
- Approves the user requirement specification (URS), that defines what the users require the system to do for them
- Approves the validation plan (VP), that defines how the validation will be undertaken
- Approves the functional risk assessment (FRA) that defines the systems functionality that will meet the URS. The risk of each function will determine the level of testing for that function, more risk involves a higher level of testing. Must be able to trace how each user requirement is qualified
- Approves testing strategy and summary reports for the operational and performance testing (OQ/PQ)
- Approves the validation report (VR) that defines proactively if the system has passed validation and can be used
- Assures compliance with defined policies and procedures.

Where vendor documentation is leveraged in support of validation, QA ensures the quality of documentation is comparable to internal standards and that the vendor supplied documents are approved internally; this can be achieved by the addition of a signed cover sheet to the validation deliverable.

Post Validation Activities of QA

The GxP computerised system must be managed according to a controlled life cycle which covers change control, configuration management and problem management. QA has oversight of all life cycle deliverables, approves changes with GxP impact and monitors non GxP changes. QA will assure the site computerised systems inventory and periodic review of the validation status of computerised systems. QA approves and controls computer system SOPs and routinely audits the computer facility.

Summary

Computerised systems are an integral part of veterinary clinical, pre-clinical and pharmacovigilance environments in which QA has an integral role in the systems life cycle to ensure the integrity of electronic data. It is a challenging area to some veterinary QA professionals. However, the electronic age is truly upon us and in order to provide adequate QA overview, QA professionals must understand their roles, QA oversee compliance in purchase, validation, change management, in on-going review and finally in computer system retirement.

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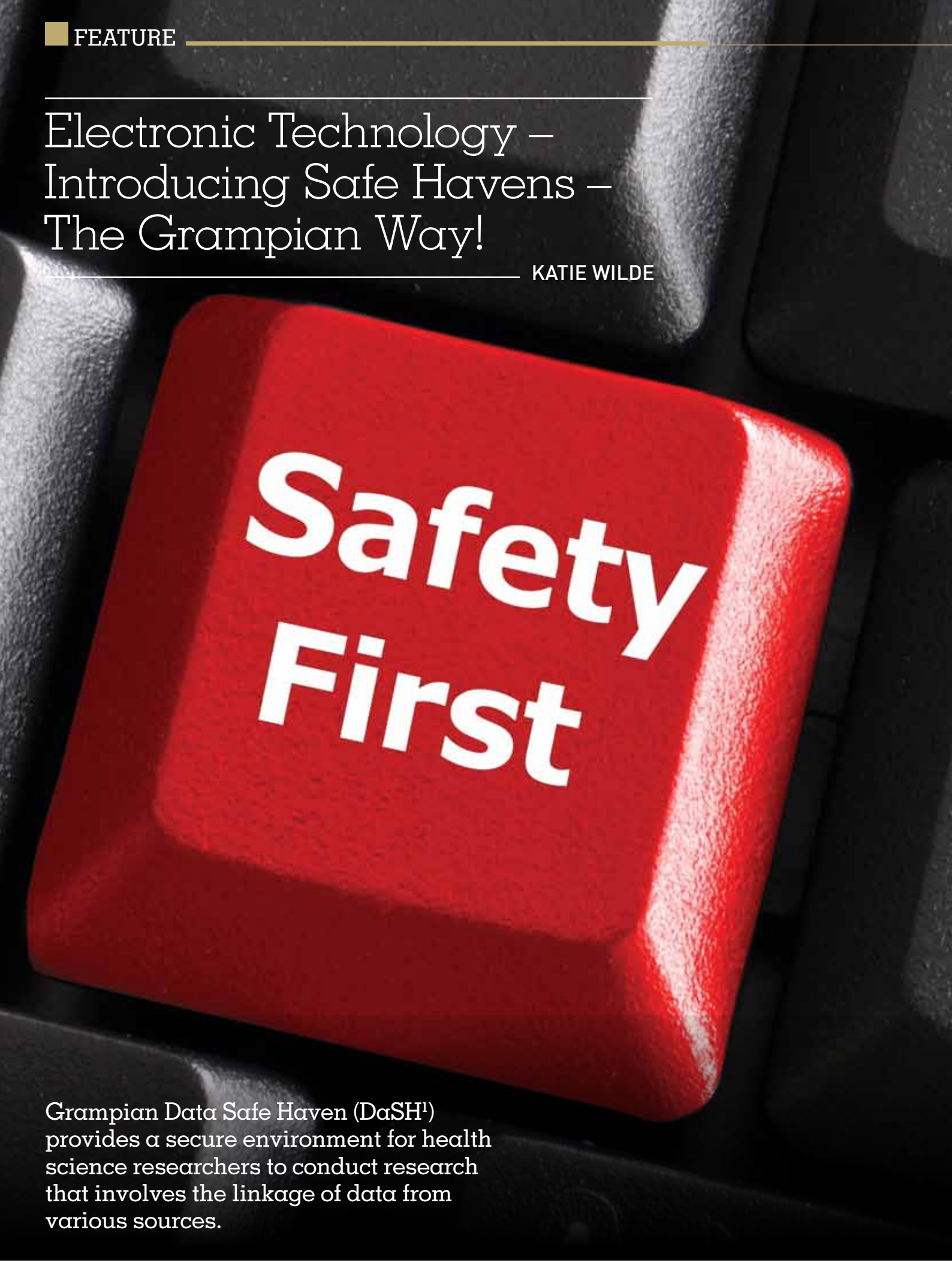
IAIN MCPHEE

From a nutrition/biochemistry background as Study Director developing trace element medications for ruminants, Iain became involved in quality with the EU requirement for GLP in veterinary pre-clinical safety studies. The facility in which he became quality manager was the first exclusively veterinary medicinal development laboratory in the UK GLP compliance programme.

Iain was a member of QAG, the precursor to RQA and a founder member of the RQA Scottish and Northern Regional Group. Iain is a past chair of the Animal Health Committee and RQA Board member. He has been on a number of committees both in RQA and allied areas to promote veterinary research quality assurance and is still an active RQA Animal Health Committee member. Currently working for Novartis Animal Health as Global R&D Compliance Manager he is concerned with veterinary GLP, GCP and GMP as applied to investigational veterinary products. He was involved in the globalisation of VICH GCP throughout Novartis Animal Health.

Electronic Technology – Introducing Safe Havens – The Grampian Way!

KATIE WILDE



Grampian Data Safe Haven (DaSH[1]) provides a secure environment for health science researchers to conduct research that involves the linkage of data from various sources.

DaSH was opened in May 2012 by NHS Grampian and the University of Aberdeen's College of Life Science and Medicine, as a joint NHS and academic facility, set up as one of the regional hubs of the Scottish Safe Haven Network.

In line with the emerging national guidelines[2] DaSH facilitates and supports safe settings, projects, people and outputs which in turn enable and create better opportunities for researchers to undertake innovative data linkage research.

Safe Settings

A preconception of a 'safe haven' is a secure, locked room for which you have to book time, to ensure that all researchers across the institutions at different campuses can access this service. DaSH has gone down the virtual access route. This enables the research team to logon to a secure virtual private network (VPN) to access their data securely but without the ability to 'print' or 'copy' so that the data does not leave the safe haven. This is a better fit with the requirements of the DaSH users in that it does not require rigid booking systems, avoids often lengthy travel time and also the possibility that the researcher only realises, when they look at the data, that what they had planned to do is going to take a lot longer than the afternoon they had scheduled.

Behind the scenes DaSH comprises of state-of-the-art dedicated computer servers to safely store patient data. Identifiable data, crucial to enabling linkage of different datasets together is stored within a dedicated NHS DaSH server, to the highest levels of security, whilst the VPN and data storage centre sit in the newly refurbished University of Aberdeen Data Centre. Access to these servers is controlled by the respective IT Department; all users of the NHS server hold a NHS contract, an honorary contract or a research passport.

Safe Projects

Research coordinators are involved in the project from the outset. They work closely with the researcher and data management team to ensure that the handling of public data are carefully planned, well documented and securely managed for research and archiving. Each project has a data management plan which defines how the data will be handled, stored, archived and deleted; and a data linkage plan which defines where the data is coming from, when it will be linked, which developer will be doing the linkage, where it will be stored and which variables are potentially identifiable and therefore removed.

One developer is assigned the role of extracting the cohort from the various research datasets that the university holds, they will then assign a pseudo-anonyimised link number which is also added to the details stored on the NHS computer. This allows us to re-check the linkage or re-link in future.

This is then sent to the relevant department, in the majority of cases, NHS National Services Scotland Information Services Division (ISD), who then extract the data required such as hospital admissions and returns this file without the identifiable information, just the link number. This is sent to a second developer who is not privy to the identifiable information. They then perform any additional linkage and a sense test before moving it in to the dedicated research area.

A sense test is performed before data release to allow for any possible mismatch in linkage to be identified. Each test is different and is dependent on the particular data source and the type of data used. Examples include checking that if someone has a date of death, there are no subsequent hospital visits. We also use this test to highlight any disease specific groups with small numbers of people. Expert advice is sought to determine whether there is a potential risk to patient confidentiality in the event that the disease is rare or uncommon.

Safe People

Often the weakness in information security is not in the technology, but those that use it. DaSH ensures that users of our services are adequately trained. There are currently two courses available which confer Approved Researcher Status. Researchers are also expected to hold a valid GCP Certificate and sign a Memorandum of Understanding with respect to data storage, use and protections of data subjects. One of the biggest challenges is ensuring that users adopt the safe haven ethos.

The DaSH team hold GCP and Research Approved Status and are audited internally on an annual basis. A dedicated member of staff is responsible for the quality assurance of each project. Their remit is to ensure that each project has all the relevant paperwork completed and that the data is processed in line with the data management plan and DaSH working protocols. The working protocols cover areas of security and archiving and are reviewed biannually.

Safe Outputs

A statistical disclosure check, performed by an independent senior statistician, is carried out prior to the data being released to the researcher. This is to ensure that any data manipulation has not resulted in the possible identification of a patient. The named, approved researchers on a project can then safely access their data for analysis without the need to transfer data outside the safe haven avoiding any possibility of 'a memory stick being left on a train' scenario.

The DaSH approach ensures that researchers responsible for the custodianship of a dataset can retain rigorous control over who has access to their research data and a researcher cannot pass on the data without prior approvals. This is crucial to ensure that data is not subsequently combined with other data sources resulting in possible identification of individual patients

Summary

DaSH offers the highest level of security and good governance processes to protect patient confidentiality whilst at the same time facilitating high quality data linkage research studies which will influence policy and practice to improve healthcare and public health, nationally.

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KATIE WILDE

Katie is Research Applications Manager and DaSH Data Management and Technical Lead. She joined in the University of Aberdeen in 2007 after graduating from the University of Exeter. In 2010, Katie took the lead of the Data Management Team, a group of dedicated programmers assisting researchers with their healthcare related research and is responsible for the DaSH data management and technical requirements.

Electronic Source Documentation – Current Regulatory Expectations and Challenges for Industry

JON READ



The FDA released draft guidance on 'Electronic Source Data in Clinical Investigations' in November 2012.

Draft guidance on this topic was previously published by the FDA in December 2010 which has been revised in response to feedback from industry. This article will highlight the key changes between the 2010 and 2012 FDA guidance documents, draw comparisons with the EMA reflection paper on expectations for electronic source data which became effective in August 2010 and highlight some of the challenges faced by industry in complying with the expectations of regulatory agencies.

The FDA draft guidance document of December 2010 was built around the theme of three tiers of data management (Tier 1 – Data Entry, Tier 2 – Data Review, Tier 3 – Data Processing and Transmission). There were concerns from industry that the guidance was too restrictive by implying that only in exceptional circumstances could data be forwarded to parties (sponsor, contract research organisation (CRO), data management centre) at Tier 3, i.e. before investigator sign off of the electronic case report form (eCRF). This would therefore negate one of the real advantages of the eCRF, i.e. 'real time' review of data. The new guidance moves to dispel this by encouraging viewing of data by sponsors, CROs and data safety monitoring boards before investigator sign-off.

Concerns were raised over the scope of the initial FDA guidance. Although titled 'Electronic Source Documentation in Clinical Trials', the focus was on the flow of data into the eCRF. Despite the removal of the three tier theme from the updated guidance, it still does not address in detail the requirements for electronic source documents as it instead concentrates upon eCRF data entry, review and access. There is a need for further guidance on what are acceptable standards and processes to be used for the creation, maintenance, monitoring and auditing of electronic source documents in clinical trials.

One of the main concerns of the initial FDA draft guidance was that it would place additional burden on clinical investigators as for example, the generation of their own write-protected copy of the eCRF. This requirement has now been removed and the new guidance merely states that access to a signed copy of the eCRF should be controlled by the investigator and made available upon request during an inspection. The previous guidance also stated that review and sign off on completed portions of the eCRF was required prior to release of data to the sponsor, CRO, institutional review boards, etc. The concept of completed portions being reviewed and signed before release has been removed. However, there is still an expectation for periodic review and electronic signing of the eCRF by the investigator.

Standard current practice is for the investigator to sign the eCRF after a subject completes the study. This requirement would therefore increase the investigator's workload but would perhaps provide the sponsor with greater assurance that the investigator has ensured the accuracy and completeness of the data reported, as required by ICH GCP 4.9.1.

Validation

A significant difference between the FDA and EMA papers is the agencies' expectations for validation. There is no clear requirement in the FDA guidance. The 2010 guidance stated that the 'FDA may ask the sponsor and/or investigator during an inspection for information on the reliability and integrity of the software and equipment used to record and transmit data elements'. This has been removed from the 2012 guidance. The EMA paper is clearer; it states that records of system validation (with a lengthy list of deliverables) should be maintained. Bearing in mind that electronic health record systems (EHRS) are expected to meet all requirements of the paper records, this could be a challenge for investigators and software vendors. From experience, records of testing and release of EHRS is limited (if indeed they exist at all!). What is unclear is the EMA inspectors' expectations; for example, would they accept a certificate of release from a software vendor or is user acceptance testing a requirement?

Challenges for Industry

An expectation of the EMA paper is that 'the sponsor must assess EHRS in use to determine how well they meet the requirements of GCP including those detailed in the paper'. This task will usually be conducted by the monitors so it is important that they are fully trained on the requirements and providing a checklist would be a useful tool during a pre-study site visit. Selecting the right person to talk to when making this assessment is also important; the investigator may not be able to answer questions on a system's security, access and audit trail functionalities and may therefore require support from IT colleagues or even the software vendor. Functionality of some older systems may make it more difficult to comply with current expectations although the EMA paper does allow for workarounds. For example, the requirement for secure, computer-generated, time stamped audit trails can be replaced by alternative methods that fulfil audit trail requirements, such as signed and dated printouts which would enable the full reconstruction of events.

Gaining access to EHRS can prove to be a challenge for monitors and auditors alike. Firstly, some institutions can be reluctant to grant access; secondly, some systems are not sophisticated enough to restrict access to study subjects only which raises data confidentiality issues. If this is the case, copies from the EHRS should be made and provided to the monitors and auditors for source data verification. Of course, these copies should be verified by a member of the study team as a complete and accurate copy of the record on the EHRS. Monitors should perform a check of a sample of the printouts against the electronic records, under the supervision of the site staff, to ensure they are complete and accurate. Discrepancies between paper copies and the electronic record are a common audit finding.

It is important to bear in mind that both the EMA and FDA documents are guidances rather than regulations (and that the FDA guidance is still in draft stage and therefore subject to change!). Caution should be used when applying the principles as there is always the danger of over-interpretation of new legislation as it could be argued, was the case with 21 CFR part 11. A question is raised as to whether 21 CFR part 11 will be considered applicable to electronic source records and EHRS? The FDA has previously implied that this would not be the case for EHRS but the mere fact that 21 CFR part 11 is referenced in the FDA guidance could suggest a changing of opinion.



JON READ

Jon works within Gilead Science's Regulatory Compliance department. In his role, he works with a team of compliance specialists in the UK and US conducting and managing a range of GCP/GLP/GPvP audits, preparing for and facilitating regulatory agency inspections and providing training/consultancy to business groups.

Prior to joining Gilead, Jon worked as a GLP Auditor for 10 years; working in both large pharma and CRO.

Jon has been an active member of RQA for a number of years; as well as sitting on the Publications Committee, he has presented GCP regulatory updates at RQA regional forums.

VP
VIEW POINT



QA Programmes and Service Provider Assessment

It was recently reported by The Food Safety Authority of Ireland (FSAI – 15th January 2013) that a targeted sample of meat products destined for UK supermarkets was found to be contaminated[1].



This was not only shocking, it also caused a storm of concern by all vested parties. Although it was reported there were no health or safety issues involved, the products in question nevertheless contained ingredients that were not supposed to be there. In the worst case, beef burgers, intended for some major supermarket brands, were found to contain 29% equine meat. Furthermore, of the 27 beef burger products analysed, 10 tested positive for horse DNA with 23 testing positive for pig DNA. More recent analyses of samples taken from other supplies of ready meal beef products have found up to 100% contamination.

While it was reassuring to hear that none of the products identified posed any food safety risk and consumers should not be worried, it was the FSAI that provided this reassurance, rather than the supermarket itself. It would appear that the latter, under the prevailing circumstances, was incapable of providing an accurate statement of content and safety. If the supermarket didn't know what is in it, can there be any confidence in the claims at the retail stage that the food had been tested and found to be safe?

These findings should cause great worry for the supermarket senior management and begs the question did they not have a QA policy dictating the need to perform routine QA audit or assessment of the source suppliers? The meat processing company in the meantime is to halt all production at its Irish facility and announced that it was sending two auditors to conduct 'unannounced spot checks' at facilities belonging to two continental suppliers. Since the initial news story broke, it would now appear that the problem is more widespread across mainland Europe with suspicion being directed to abattoirs in both Poland and Romania and various distribution and storage facilities.

Anyone with even a modest understanding of quality and safety applied to industries as diverse as food, pharmaceuticals, medicine, engineering and technology would be only too aware of the importance of effective QA programmes in business operations and manufacturing. Without an adequate and effective QA programme operating through all stages of development and the life of manufacture, how can there be any claims for the product achieving its design specification? Quality must never be compromised. But unfortunately there is always the temptation to ignore the important detail of actually visiting the service supplier and subjecting procedures, systems, facilities and current performance to thorough examination. In this example it may be that the supermarket decided to delegate these important aspects of QA to the supplier. But a reliance on delegating QA functions, especially to a supplier, is fraught with huge danger unless very carefully thought through beforehand.

Approved and accredited facilities are no guarantee of quality output either. The meat processing company's stated policy of only buying from licensed and approved EU suppliers unfortunately doesn't appear to have provided any guarantee of the quality of the finished material in the case of the sample tested by the FSAI. While in the case of clinical laboratories, inadequate systems and equipment can be readily found despite the facility being accredited. Over-reliance on accreditation certificates during clinical trial set-up and routine monitoring can lead to similar problems as found by many QA professionals, as such accreditation is often found to be poorly representative of what the facility is actually achieving in practice on a day-by-day basis.



Looking at the problems recently in the news concerning modern passenger aircraft manufacture; here is another case of a well-known company with an impressive reputation, selling a finished product assembled from components bought in from many and varied source suppliers[2]. Huge investment in the design, development and manufacture of the aircraft is reported as having been compromised by the inclusion of two lithium ion batteries supplied from a small company in Japan. Several safety incidents associated with these batteries on operational planes have resulted in the issuance of an emergency grounding directive by America's Federal Aviation Administration (FAA). The buying of components from other companies should not automatically lead to handing all responsibility for ensuring the quality of the component to the supplier as the ultimate responsibility for quality shouldn't be delegated down. Consideration should also be given to whether any bias is likely to influence compliance with standards. In this case, the aircraft manufacturer cannot ignore its responsibilities during development and manufacture, as all aspects of quality ultimately reside with them.

Similarly, in the world of medicines, the research, development, manufacture and packaging of drugs is the responsibility of pharmaceutical companies. At the R&D stages, it has become common practice for pharmaceutical companies to hire the services of clinical research organisations (CRO), in order to either fill a gap in a project's development programme or, in some cases, to take full responsibility for the management of the project. Regardless of the size of the companies concerned, it is always the pharmaceutical company that should take ultimate responsibility for the project and all stages of the development. In the case of a small biotech company for example, even if the CRO has a very experienced and well qualified medical expert, the client pharmaceutical company should still ensure that it has a well qualified and experienced medical professional to take responsibility for decisions taken on its behalf. A definitive listing of all duties showing responsibility-ownership should be drawn up between the two parties and made official by inclusion of dated signatures of the two respective senior executives and countersigned by a senior QA professional.

Abdication of responsibility for assuring quality of services and finished products, with the resulting possible discovery of deficiencies by government agencies, is unwise and could prove very expensive in the long term, especially if it is decided in the case of pharmaceutical companies, that they can no longer be trusted.

NIGEL CROSSLAND

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RQA Annual General Meeting 2013



The RQA Annual General Meeting will be held on the 6th November 2013 at the National Motorcycle Museum, Solihull, West Midlands.

The meeting will take place at 12.00hrs.

All RQA Members are welcome.

The AGM will be held in conjunction with a number of seminars at the same location, thereby allowing those eligible seminar delegates to attend the AGM also.

Seminars are currently being developed by the following RQA Committees:

- Good Laboratory Practice
- Computing – 'Electronic Quality Management Systems'
- Medical Devices – 'Medical Device Software'
- Good Clinical Practice/Pharmacovigilance.

Details on these seminars together with their programmes will be available in due course.

Organisation Wide Quality

First Meeting of Special Interest Group
Madingley Hall, 11th February 2013

Following the two articles in the January 2012 edition of Quasar (on Root Cause Analysis and Complexity Management) and the collection of members' views for the 'Organisation Wide Quality' poster at the 2012 Annual Conference, a group of members attended an RQA meeting at Madingley Hall to explore the possible benefits in considering quality in its wider organisational context both for its members (and their employers and clients) and for the RQA itself.

Specific purposes were to share experiences and tools that could help RQA members broaden their 'Quality Management Role' and if appropriate, agree some next steps to develop the ideas and take them forward.

In Discussion

The group agreed that the secret of getting things to 'work' lies in the connections between them; that 'audit' is an assessment (similar to marking an exam paper at the end of a course) and it's the connections (often informal) that actually get things done. The wider role (or 'quality hat') helps people and organisations to learn and to deliver value.

This group will develop ways for quality professionals to 'elbow out' – to build a wider role with space to offer more than just audit, offering greater efficiency, effectiveness, potential and higher performance rather than dealing with the same things as they come around again and again.

The wider approaches discussed included the Deming Cycle (Plan–Do–Study–Act) - developing knowledge and learning through experiment. Also: corrective and preventive action (CAPA), total quality management (TQM), risk management and disaster recovery planning and the dynamic organisation (for details see below on how to obtain the poster) and the common threads in GxP and PV.

- How do we decide what to measure?
- Asking the right questions is very powerful.

The RQA might consider some of these ideas as it realigns its developing role (and identity) as implied in its new name 'Research Quality Association' (is it to be 'compliance' or 'quality'). Implications for its ongoing financial health were also considered.

Review of the Day - Insights/Issues and Opportunities:

- As fashions come and go, the underlying need remains
- Similar to the impact of TQM (both internal and external 'customers' and 'suppliers')
- It's about building common thinking throughout organisation, and moving from 'criteria' to address wider expectations, beyond audit
- Different models to include (so far) Deming, CAPA, TQM, SCiO Organisational Maturity Model (OMM)
- 'Like' the simplicity of Dynamic Organisation Model.

Next Steps, it was agreed:

- There is sufficient energy to take these ideas forward
- To update those who have expressed interest but were unable to attend the meeting
- To encourage others to join the group (either to take part or to be regularly updated)
- To build an agenda for next meeting.

Joining the Group

If you would like a copy of the poster presented at the 2012 Annual Conference, or are interested in being put on the circulation list for this group please contact: **Dave Mettam**

Telephone: 0161 443 1420 Email:dave@interact-development.com
copy to Dave Weller **dweller@therqa.com** at the RQA Office.

UPDATE

Computing

LIZ ADAMS
COMPUTING COMMITTEE SECRETARY

Committee Activities

The last meeting was held in January at GE Healthcare, Amersham. Andy Ward has stepped down as Committee Secretary and Liz Adams has volunteered to take on this role. All of the Committee expressed appreciation to Andy Ward for delivery of outstanding commitment and service for numerous years as secretary.

Two potential new members, Susan Widlake and Matt Davies, joined the meeting and have been accepted as Computing Committee members.

The Committee discussed the programme for the first European QA Conference to be held in September. A number of potential topics have been suggested for the computing stream.

Other areas discussed included:

- Status of current articles in preparation and hot topics
- Interaction with other committees and societies
- CSV in China and the future of computerised systems
- Electronic Source Data in Clinical Investigations – FDA draft guidance
- Global SQA Conference, 2014.

Professional Development Courses

The next computing professional development course to be run is:

- Auditing Computerised Systems, 3-5th June 2013, Madingley Hall.

Booklets

Don't forget the recently published booklet 'Auditing Computerised Systems'. For details see the article in the last edition of Quasar. The 'Computerised System Validation: Risks, Requirements, Tests and Traceability' booklet has also been updated; both these booklets are available on the RQA website and are in support of the professional development courses.

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www.therqa.com

Questions and Answers

Some recent computing related issues from the RQA discussion forum are included below.

Question: We are in the process of implementing a commercial tool to manage our audit reports which is fully audit trailed. We're having an ongoing debate within the QA team as to whether it is necessary for the auditor to sign an audit report? We cannot identify a regulatory requirement preventing us from eliminating this practice. Has anyone had any experience of removing the signature requirement from their audit reports? Would you have any objections if you saw this practice at a CRO?

Computing Committee: Are you asking whether an audit trail is equivalent to an electronic signature? If so I would refer you to FDA 21 CFR Part 11, eRecords, eSignatures for clarification. The regulatory authorities require that when an electronic signature is used in a document it is the legally binding equivalent of a paper signature. There are strict requirements regarding eSignature. In our experience a regulator has never indicated that no signature at all is equivalent to a paper signature. Have you asked your clients for their views about this?

Question: We are building a genomics facility. The equipment generates raw data files that are 4 Tbytes each, after manipulation they are a fraction of this size. Would we be in compliance if we only saved and archived the manipulated (smaller) files?

Computing Committee: Compliance nowadays is related to perceived risk to patient safety, product quality and data integrity, so it will depend on what you are using the data for. In GLP it is necessary to define precisely what the raw (source) data is and to be able to reconstruct the study from the archive. GCP also has requirements around source data (see ALCOA guidance). Product-related data from clinical studies is held in archives for decades. On the other hand the Data Protection Act requires that the minimum amount of personal data is held and only for as long as required. You should consider business risk too when deleting files in terms of regenerating them should you need them.

GCP Committee: Provided there is full traceability of data points and the owner can verify traceability, then this would be deemed acceptable. The process for data management and archiving should be fully defined so that the files can be reconstructed for audit and inspection purposes. There should be provision for security and access permission to ensure the integrity is kept.

Reference should also be clearly made to having a full quality control/validated process for the 'manipulations'.

UPDATE

Good Clinical Practice

ANGELIKA TILLMANN
GOOD CLINICAL PRACTICE COMMITTEE SECRETARY

Committee Activities

In the first quarter of 2013, the GCP Committee organised two telephone conferences and one face to face meeting. One seminar was organised by the GCP Committee. The MHRA published their 'GCP Guide' in September 2012, the RQA seminar on implementation and impact of the Guide was held on 22nd January 2013 at the Novotel, London Heathrow with Colin Wilsher as Seminar Principal and Patricia Henley and Kath Williams as tutors. Gail Francis from the MHRA opened the seminar which was fully booked and received very high feedback scores.

Following the success of the one-day seminar we have now produced a short (62 minute) webinar based on the seminar. The webinar concentrates on what elements of the GCP Guide appear to be new or different from the perspective of a GCP Auditor and what opportunities it might offer. As the GCP Guide is 542 pages long, any short review will, by definition, not be comprehensive. However, it is hoped that the webinar will allow an appreciation of what GCP challenges and opportunities the new MHRA GCP Guide may provide. This webinar is conducted by Colin Wilsher and takes a high level view, it is assumed that attendees have detailed knowledge of ICH GCP E6, the EU Clinical Trials Directive, the EU GCP Directive and the relevant UK Statutory Instruments (SIs). It is available for purchase on the website now.

The RQA course 'Auditing Electronic Data Capture' was held on 13th-14th March 2013 by two members of RQA's Computing Committee, Thana Subramanian and Matthew Jones and two members of the GCP Committee (Keith Miller and Angelika Tillmann, who is also the Course Principal).

The preparations for the first European QA Conference in September 2013 in Bonn, Germany are well under way. This conference is organised by the QA associations from France (SOFAQ), Germany (DGGF) and the RQA. Angelika Tillmann, the GCP Committee representative on the Programme Committee attended the face to face meeting in London in January and is also attending the regular conference calls. The programme is almost finalised and several regulatory agencies are going to send speakers for this international conference.

The GCP Discussion Forum on the RQA website is continuously monitored by Colin Wilsher and some of the more complex questions have been answered by members of the GCP Committee and are published in this edition of Quasar and the Q&A section of the website.

The GCP Committee is also monitoring any regulatory changes and has provided comments to the FDA via EQAC on the draft FDA 'Draft Guidance on Electronic Source Data in Clinical Investigations'. Additionally, there is a summary on the very recent and major changes to regulations in India.

GCP Committee News

The GCP Committee has been able to fill it's vacancy with Robrecht Tistaert. The GCP Committee members are looking forward to working with him.

Regulatory News

The MHRA published an update on the 'Serious Breach (SB) Guidance' on 7th February 2013. Changes include that National Research Ethics Service (NRES) standard operating procedures require notification of SBs and that in the updated guidance, the option of faxing or mailing in SB forms to the MHRA has been removed.

On 22nd January 2013, the EMA published the 'Technical Guidance on the Format of the Data Fields of Result-related Information on Clinical Trials'. In its guidance 2012/C302/03 on posting and publication of result-related information on clinical trials in relation to the implementation of Article 57(2) of Regulation (EC) No 726/2004 and Article 41(2) of Regulation (EC) No 1901/2006 the commission has announced, under point 3, publication of implementing technical guidance on the format of the data fields in relation to result-related information.

This newly published document contains information on implementing technical guidance. It provides a visual representation of the clinical trial results data that is required to be captured by EudraCT.

It also provides details of how the fields are organised. Results may be displayed in the EU clinical trials register in a different visual representation. This document does not describe the user interface design for entry of results data in EudraCT.

For additional regulatory information, reference is made to the commission guidance 2012/C302/03. Technical information on EudraCT is available at https://eudract.ema.europa.eu/index.html.

The consultation period to the EMA regulation on advanced therapy medicinal products was open until 31st March 2013 and stakeholders were invited to provide comments.

As mentioned, in India the clinical trial regulations governing the conduct of clinical trials were amended in January and February 2013, following long discussion and many publications about exploitation of clinical trial subjects. Please refer to the separate summary on the main changes.

Questions and Answers

Ethics Committee Member Conflict of Interest

PREPARED BY MARTHA BYRNE

Question: I would like to seek your opinion on a conflict of interest in the following scenarios:

1 - The authorised signatory on the contract for the site (owned by the individual) which includes the budget as well, is a member of the Ethics Committee who had opined on the conduct of the trial.
2 - If the institute is not owned by an individual person, yet the authorised signatory on contract (someone appointed by the institute) signs the contract and also is a member of the Ethics Committee which had opined on the conduct of the trial.

GCP Committee Response: It looks like a conflict of interest as the person on the Ethics Committee cannot provide an impartial decision. There is a Q&A from the Committee on the website concerning conflict of interest.

If the Ethics Committee member signs a contract on behalf of a site, they retain responsibility for contractual arrangements and there is a direct conflict of interest if the individual then approves research as they are not completely independent. If this query is being done prospectively, then in such instances, the Ethics Committee Member has the opportunity to declare a conflict of interest and not participate in the decision making of the Ethics Committee on this application. If this has already occurred, then a declared interest must be notified and confirmation requested on (1) the scope of the opinion in relation to the conflicted Ethics Committee Member, (2) seek clarification on whether the involvement of the conflicted member had any direct bearing on the outcome/approval of the proposed research and document the proposed corrective action to demonstrate the independence of Ethics Committees in their review of research.

Removing Source Documentation from an Investigator Site

PREPARED BY COLIN WILSHER AND ROBRECHT TISTAERT

Question: What is your opinion about taking source documentation of patients from an investigational site to an external location (e.g. hotel room or home) for monitoring using the remote data capture system? Would it be a GCP violation?

GCP Committee Response: Generally, the answer is 'no' to taking (copies or) original source documents from investigator sites, as the process is risky from a GCP and data privacy perspective. According to GCP, source documents must remain at the investigator site (ICH E6 section 8; 2.11; 4.9.4; 5.18). Also the EMA inspectors (refer to the EMA website and reflection paper and CDISC guidance) and the FDA are very concerned that the sponsor (or the sponsor's agent, such as the monitor) should not have exclusive control of the investigator's copy of the data. As you explain that it would be the monitor taking these documents, then this would not be allowed. However, it might be possible for a member of the investigator site to remain with the documents at all times and hence they would be under the control of the investigator and his team.

Local privacy regulations may be specific regarding not allowing patient identifiable data to leave the hospital and the privacy language of some consent forms should be carefully reviewed. Sometimes, the consent form states that reviews of health records will only occur at the research site.

In addition, you should be aware of hospital/institution directives that either the person taking the source and/or the person allowing the source to be taken may be violating.

Last but not least, by removing source documentation from an investigator site, you take on a huge responsibility that may not be in compliance with or supported by your company policies.

In specific situations, regulatory authority inspectors have requested copies of informed consent forms and other source documents. It will be important to ensure that data privacy legislation is complied with when providing this information.



Summary of Recent Changes in Clinical Trial Regulations in India

ANGELIKA TILLMANN

The following summary is intended to provide an update on the main changes in India. For full information, please refer to the links provided at the end.

The clinical trial regulations governing the conduct of clinical trials in India were amended in January and February 2013, following long discussions and many publications in India the about exploitation of clinical trial subjects. These events have occurred in parallel to a series of unannounced inspections of investigational trial sites.

One amendment to the regulations was published on 30th January 2013 and describes the requirement to provide medical treatment for any injury occurring during a clinical trial, regardless of whether the injury is related to the investigational product/device or not. The amendment further requires that compensation is to be paid to the subject or heirs of the subject for all serious adverse events (SAEs) and deaths, as described in number 5 below.

The following paragraphs are an excerpt of the amendment (compensation regulation):

1. In the case of an injury occurring to the clinical trial subject, he or she shall be given free medical management as long as required.
2. In case the injury occurring to the trial subject is related to the clinical trial, such subject shall also be entitled for financial compensation as per order of the licensing authority defined under clause (b) of rule 21, and the financial compensation will be over and above any expenses incurred on the medical management of the subject.
3. In the case of clinical trial related death of the subject, his/her nominee(s) would be entitled for financial compensation, as per the order of the licensing authority defined under clause (b) of rule 21, and the financial compensation will be over and above any expenses incurred on the medical management of such subject.
4. The expenses on medical management and financial compensation in the case of clinical trial injury or death of the trial subject shall be borne by the sponsor of the clinical trial.
5. Any injury or death of the subject occurring in clinical trials due to following reasons shall be considered as clinical trial related injury or death and the subject or his/her nominee(s), as the case may be, are entitled for financial compensation for such injury or death:
   a. Adverse effect of investigational product(s)
   b. Violation of the approved protocol, scientific misconduct or negligence by the sponsor or his representative or the investigator
   c. Failure of investigational product to provide intended therapeutic effect
   d. Use of placebo in a placebo-controlled trial
   e. Adverse effects due to concomitant medication excluding standard care, necessitated as part of approved protocol
   f. For injury to a child in-utero because of the participation of parent in clinical trial
   g. Any clinical trial procedures involved in the study.
6. The sponsor, whether a pharmaceutical company or an institution, shall give an undertaking along with the application for clinical trial permission to the licensing authority defined in clause (b) of Rule 21, to provide compensation in the case of clinical trial related injury or death for which subjects are entitled to compensation.
7. In case the sponsor fails to provide medical management for the injury to the subject and/or financial compensation to the trial subject for clinical trial related injury or financial compensation to the subject's nominee(s) in case of clinical trial related death of the subject, the licensing authority may after giving an opportunity to show cause why such an order should not be passed, by an order in writing, stating the reasons thereof, suspend or cancel the clinical trial and/or restrict sponsor including his representative(s) to conduct any further clinical trials in the country or take any other action deemed fit under the rules.

Based on the above conditions implemented by the new amendment to the existing regulations, the company submitting the clinical trial application and/or acting as legal representative in India is required to pay for the medical treatment and compensation. It is possible for a CRO to include indemnification via contracts with the sponsor. However the risks and associated costs for the conduct of full service trials are currently not foreseeable.

The reporting requirements for SAE and deaths have also been amended. The investigator has to initially report all serious and unexpected events within 24 hours of occurrence to the sponsor/CRO and Ethics Committee, and has to submit a full report to the Ethics Committee within 10 calendar days. This new SAE reporting requirement is stricter than other global regulatory requirements which request reporting within 24 hours after investigator awareness of the SAE to the sponsor/CRO. The investigator is depending on patient reporting and therefore might not be able to report every SAE within 24 hours of occurrence.

All SAEs including deaths are further to be reported to a newly formed Expert Committee for the evaluation of the SAE/death for the determination of causality of the death and the quantum compensation.

In addition, another regulatory requirement related to SAE including deaths reporting is in the process of implementation. A system of pre-screening of SAE reports for submission to the Ethics Committees and the newly formed Expert Committee was implemented on 18th February 2013. This system of preliminary screening to determine the acceptability of the SAE report is meant to pre-screen the SAE to be reported under the new compensation regulation. As per those provisions, each SAE including death is required to be examined for a decision regarding causality of death and quantum of compensation.

It is important to note that failure to comply with the new requirements may lead to the clinical trial approval being revoked for the study where the non-compliance occurred and all other studies conducted by the same investigator/sponsor/CRO.

Another amendment dated 1st February 2013 was implemented which is related to inspections and describes additional conditions for clinical trial conduct in India.

The following text is an excerpt of the second amendment:

- Premises of sponsor including their employees, subsidiaries and branches, their agents, contractors, sub-contractors, clinical trial site shall be open to inspection by officers authorised by Central Drugs Standard Control Organisation (CDSCO) who may be accompanied by officers of state drug control authority
- Above parties shall allow (CDSCO and State Drug) officers to enter with or without prior notice any premises to inspect, search and seize any data, records, documents, books, investigational drugs related to clinical trial and provide adequate replies to queries raised by inspecting authority
- Licensing authority may also impose additional conditions for issuance of permission, if considered necessary regarding the objective, design, study population, subject eligibility, assessments, conduct and treatment of clinical trials
- If parties mentioned above fail to comply with the above conditions, the licensing authority may, after giving show cause why such an order should not be passed, by an order in writing stating reasons thereof:
  - Issue warning letter giving details of deficiency found during inspection
  - Recommend that the study may be rejected or discontinued
  - Suspend or cancel clinical trial permission
  - Debar the investigator(s), sponsor including their employees, subsidiaries and branches, their agents, contractors, sub-contractors to conduct any clinical trial in future.
- Investigators against whom the action has been taken by the licensing authority, may within ninety days (90 Days) of receipt of copy order of the licensing authority prefer an appeal to the central government, and the central government, after giving such appellant an opportunity of being heard, confirm, reverse or modify such an order.

The above Drugs and Cosmetics (D&C) Rule changes necessitate reviews and revisions to current patient information sheets/informed consent forms, insurance policies and indemnification language in contracts.

After amending the D&C Rules to provide clarity to the compensation regulation and requirements of obtaining permissions for the trials, the Health Ministry has released a notification for mandatory registration of ethics committees within 45 days, effective 8th February 2013. As per notification, issued by the CDSCO '....No ethics committee shall review and accord its approval to a clinical trial protocol without prior registration of the committee with the licensing authority as defined in clause (b) of rule 21....'.

As per the above amendments to the regulations, clinical trials only should be approved by 'registered Ethics Committees'. SAE will have to be reported to and evaluated by 'SAE Expert Committees' that now need to be established following the implementation of the regulations.

Effective 25th February 2013, a system of preliminary scrutiny to determine the acceptability of the application for registration of ethics committees (ECs) will be established. CDSCO officer(s) will perform a preliminary review of the applications of ECs applying to be registered in India. The preliminary review of the application will be performed based on the newly published checklist. During the preliminary examination, the CDSCO officer(s) will scrutinise the EC application to ensure that it contains all the required administrative as well as technical information as required. If the EC applications are not submitted in accordance with the format and the newly published checklist, they will not be accepted by CDSCO for further examination. The checklist can be found here: http://cdsco.nic.in/clinical_trial.htm

Additionally, two 'Expert Committees' with five members each will be formed to formulate policy guidelines and standard operating procedures for approval of new drugs, clinical trials, banning of drugs and fixed dose combinations (FDCs).

Further, an 'Apex Committee' with five members is to be established as per order dated 6th February 2013, supervising the new clinical trial authorisations (CTA) approvals. In addition to the above and per order dated 6th February 2013, a 'Technical Committee' with 15 members will be established to supervise and monitor the conduct of clinical trials in the country, which will support the 'Apex Committee'.

As a result of the above Independent EC changes, delays are anticipated with reviewing and approving clinical trial applications and amendments to ongoing studies during the next two months.

In parallel to the newly implemented regulatory changes, the Supreme Court in India has issued a directive to perform an investigation related to deaths and compensation that occurred during clinical trials. This triggered several unannounced inspections of clinical trial sites in January and early February 2013, prior to the Supreme Court hearing. It was announced that from February 2013 onwards, the Drug Controller General of India (DCGI) will not be approving new clinical trials in India for at least two months.

More information can be found under the following links:

http://cdsco.nic.in/GSR%2053(E)%20dated%2030.01.2013.pdf
http://cdsco.nic.in/GSR%2063(E)%20dated01%20.02.2013.pdf http://cdsco.nic.in/html/G.S.R%2072(E)%20dated%2008.02.2013.pdf
http://cdsco.nic.in/System%20of%20Pre-screening%20for%20submission%20of%20reports%20of%20SAEs%20to%20CDSCO.pdf

We will continuously monitor the regulatory situation in India for regulatory changes and provide further updates.

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UPDATE

Good Manufacturing Practice

GORDON MACDONALD
GOOD MANUFACTURING PRACTICE COMMITTEE CHAIRMAN

In the last edition of Quasar, the Committee informed the GMP constituency of a meeting held between the Committee and MHRA inspectors to discuss questions presented by RQA members. While it was hoped to be able to publish the outcome of the Q&A session in this edition of Quasar, the responses are currently being reviewed and approved within MHRA. Once approved these will be made available via the members' pages of the RQA website and published in the next edition of Quasar.

The main changes that the Committee is aware of that have occurred in the EU GMP environment over the last few months are:

- Updates to Eudralex Volume 4, GMP guidelines
- Changes within the MHRA
- New GDP guideline 2013/C 68/01 published 7 March 2013 http://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=OJ:C:2013:068:0001:0014:EN:PDF

All Change: EudraLex Volume 4 (EU GMPs)

Over the last few months multiple changes have either been made, or proposed, to some of the core EU GMPs contained in EudraLex Volume 4. Some of the main changes that have been made are outlined below with an overview of some of the proposed changes.

Recent Changes

Two chapters and one annex to EudraLex Volume 4 were updated in September 2012 and came into operation on 31st January 2013.

Chapter 1 - Pharmaceutical Quality System (previously Quality Management)

This chapter has been updated (again) to incorporate the concepts and terminology described in the ICH Q10 guideline (pharmaceutical quality system). It has also been updated to clearly refer to the clinical trials authorisation, as appropriate, thus reinforcing the concept that a quality management system (QMS) is required for companies engaged in development activities as well as commercial manufacturers/marketing authorisation holders. It should also be noted that this reinforces the principle that manufacturers of investigational medicinal products need to be aware of the content of the main chapters of Part 1 of the EU GMPs and not just Annex 13. Other changes reflect recent updates in related GMPs such as the Falsified Medicines Directive (2011/62/EC) and recent changes to other EU GMP chapters and annexes.

For the first time in Part 1, the concept of change control has been introduced and includes requirements for a prospective evaluation of planned changes and their approval prior to implementation (taking into account regulatory notification and approval where required), as well as a post-implementation review – to confirm that the quality objectives were achieved and that there was no unintended deleterious impact on product quality.

There is more detail relating to the investigation and management of deviations, with the concepts of root cause analysis and appropriate corrective and preventive actions (CAPAs) being introduced.

A quality manual, or equivalent documentation, is also now required to be established and a formal link to Good Distribution Practices (GDP) has also been made.

Chapter 7 - Outsourced Activities (previously Contract Manufacture and Analysis)

One of the biggest changes, introduced in Chapter 7, is the requirement for the contract giver to include the control, monitoring and review of any outsourced activities in their pharmaceutical quality system.

The updated Chapter 7 also includes a reminder that the contract giver is ultimately responsible for those activities which they have outsourced and explicitly requires the contract giver to review and assess the records and results related to the outsourced activities, as well as ensuring that the products have been processed in accordance with GMP and the marketing authorisation.

Annex 2 – Manufacture of Biological Active Substances and Medicinal Products for Human Use.

According to the introductory text in the new version of Annex 2, *'Annex 2 of the GMP Guide has been revised as a consequence of the restructuring of the GMP Guide, new manufacturing technology and concepts, the increased breadth of biological medicinal products to include several new product types such as transgenic derived products and the Advanced Therapy Medicinal Products, (ATMPs) together with associated new legislation'* – this is something of an understatement as the revisions to Annex 2 are substantial.

Annex 2 has been extensively re-written and has been divided into two main parts:

- Part A, which contains supplementary guidance on the manufacture of biological active substances and medicinal products, from control over seed lots and cell banks through to finishing activities, and testing
- Part B, which contains further guidance on selected types of biological active substances and medicinal products.

While most of the changes are implementing principles that should be familiar to GMP QA groups, for example use of quality risk management (QRM) principles, it is strongly suggested that those members of RQA who are engaged in the manufacture and control of biological medicinal products carefully review the updated Annex 2 with their specific product in mind, making any necessary consequential changes to their quality systems and procedures.

Proposed Changes

The following proposed changes to four of the chapters in Part 1 have also been submitted for public consultation by the EU and are available for review on the internet:

Chapter 3 - Premises and Equipment

The only proposed change to Chapter 3 relates to the provision of more detailed guidance to prevent cross-contamination (Section 6), including reference to a new complementary toxicological assessment guidance; the proposed draft describes three instances when the use of dedicated facilities are required.

Chapter 5 - Production

The 'reasons for change' for this chapter include changes to Sections 17 to 20 to improve the guidance on prevention of cross-contamination and to refer to toxicological assessment guidance. This guidance is provided in the draft document 'Guideline on Setting Health Based Exposure Limits for Use in Risk Identification in the Manufacture of Different Medicinal Products in Shared Facilities' has been published alongside the changes to Chapter 3, for comment by 30th June 2013.

Changes have also been proposed in Sections 26 to 28 on the qualification of suppliers in order to reflect the legal obligation of manufacturing authorisation holders to ensure that active substances are produced in accordance with GMP; the changes include supply chain traceability.

Section 33 has been inserted to clarify and harmonise expectations of manufacturers regarding the testing of starting materials, while Section 68 introduces guidance on notification of restrictions in supply.

Chapter 6 - Quality Control

The 'reasons for change' for this chapter include the addition of a whole new section on 'Technical Transfer of Testing Methods' (Sections 37 to 41), and other items such as out of specification (OOS) results.

Chapter 8 - Complaints, Quality Defects and Product Recalls

The following 'reasons for change' relate to the proposed draft:

- To reflect QRM principles to be applied when investigating quality defects/complaints and when making decisions in relation to product recalls or other risk-mitigating actions
- To emphasise the need for the cause(s) of quality defects/complaints to be investigated and determined, and that appropriate preventative actions are put in place to guard against a recurrence of the issue
- To clarify expectations and responsibilities in relation to the reporting of quality defects to the supervisory authority.

Draft Concept Papers

Draft concept papers have also been issued for the following annexes:

- Annex 17 - Parametric Release, deadline for comments was 28th February 2013
- Annex 15 - Qualification and Validation, deadline for comments was 28th February 2013.

What Next?

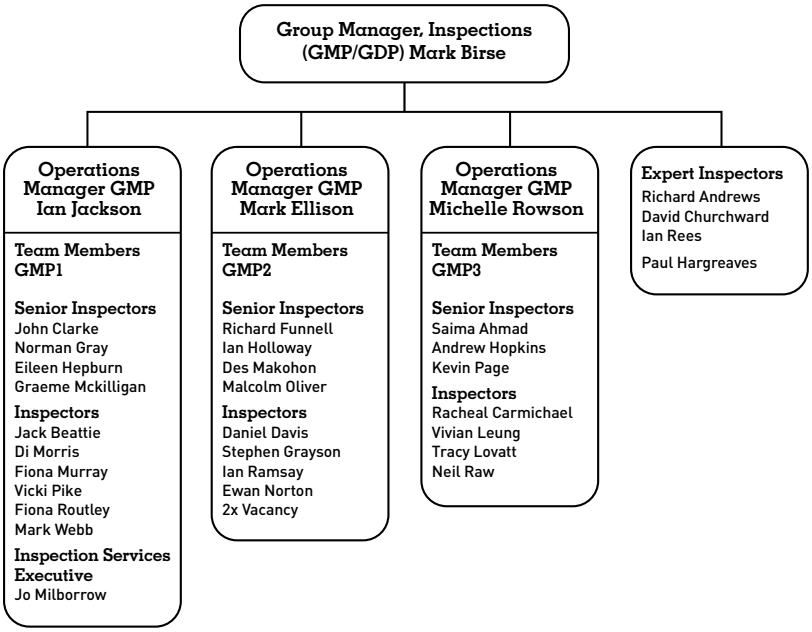
Many of the changes reflect what is current industry best practice and reflect the codifying of ICH Q9 (Quality Risk Management) and ICH Q10 (pharmaceutical quality system) principles into EU GMPs, and will not be new to GMP QA groups. However, it is recommended that you review your QMS and perform a gap analysis, to ensure that it meets all of the recently updated expectations. Also, update your standard operating procedures (SOPs) and your train your staff accordingly.

If you have any comments on any of the proposed changes which are currently out for public consultation, please get in touch with the RQA GMP Committee who will be happy to coordinate a response on behalf of all of our members and submit this to the EU for consideration.

MHRA Changes

There have been a number of changes within the GMP inspections department of the MHRA over the last year. Of note is the departure of Ian Thrussell to head up the inspections division of the World Health Organisation, the retirement of Mike Woodhall and selection of Saima Ahmad as training manager for the group.

An organisation chart derived from recent announcements and presentations on the MHRA website is below:



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UPDATE

Medical Devices

ALAN DENCH
MEDICAL DEVICES COMMITTEE CHAIRMAN

2012 Annual Conference

In the January edition of Quasar we reported on the Medical Devices QA Clinic and presentations at the 2012 Annual Conference; the Medical Devices Committee met in January 2013 to review feedback from the conference. This was rated overall as successful and the QA Clinic and sessions were particularly well received by participants. The Medical Devices Committee continues to be busy on your behalf.

Draft Regulations

Last September the European Commission released proposals for two draft European regulations covering medical devices and in vitro diagnostic devices to ultimately replace the current directives on devices, active implantable devices and in vitro diagnostic devices. There are several rationales for these proposed changes:

- European legislation on medical devices is over 20 years old and there have been significant developments in technology
- The enlargement of the EU over that time makes the existing legislative structure less tenable
- Questions have been raised about the regulatory framework for devices in Europe in the wake of recent high profile cases involving safety concerns of some metal-on-metal implants and suspected fraudulent PIP breast implants.

The main features of both proposals are:

- Increased transparency and supply chain traceability requirements
- Strengthen the designation and audit of the notified bodies responsible for pre-market safety assessment of devices
- Additional pre-market scrutiny of higher risk devices and a requirement for more clinical evidence for higher risk and implantable devices
- Establishment of central EU reporting of serious incidences and field safety corrective actions
- Improve EU member state coordination with establishment of new governance structures to align with the proposed legislative framework.

For the medical device proposal, the main changes are:

- Extension of legislative scope to include invasive or implantable devices without a medical purpose, i.e. dermal fillers and aesthetic implants.
- Introduce regulatory requirements on the reprocessing of single-use devices.

For the in vitro device (IVD) proposal, the main changes are:

- Change to a rule-driven, risk-based classification system
- Additional controls on IVD's made and used in the same health institution, with the exclusion of higher risk devices from this exemption
- Clarification that companion diagnostics and genetic tests with a medical purpose are within the scope of the regulation.

In November, the MHRA sought views and invited comments on these proposals. The Medical Devices Committee coordinated the RQA response on behalf of the Board and all committees, which was submitted as part of the European Quality Assurance Confederation (EQAC) response to the MHRA in January.

Committee Activities

The Committee recently completed a review to update the Medical Devices Regulation booklet, so that when this appears in the new RQA livery it will reflect the current status of medical devices regulation which has moved on since the booklet was first published in 2009.

The Committee is developing a medical device development roadmap which will ultimately be available on the RQA website.

The Committee is also involved in the first European QA Conference in Bonn in September, where we intend to have a session on the role of notified bodies.

And finally, in November we are planning a seminar and workshop on aspects of medical device software. More details will follow subsequently.

We are keen to engage with the membership about the future scope and direction of the Committee.

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For other committee members please see the committee pages on the website www.therqa.com

VACANCY Medical Devices



There are vacancies for new committee members, so if your interest is in the medical devices market then we would be delighted to hear from you and for you to join us.

UPDATE

Good Pharmacovigilance Practice

ALLISON JACK
GOOD PHARMACOVIGILANCE PRACTICE COMMITTEE

The Good Pharmacovigilance Practice Committee met in December 2012, and minutes of the meeting have been published, they last met in March 2013.
A number of seminars are proposed for 2013, which we hope will be of interest to the membership, specifically:

- A follow up to last year's seminar on the 'Implementation of the New PV Legislation' (planned for the second quarter)
- 'Medical Devices/Device Vigilance' (planned for the second quarter)
- A follow up to the previous seminar 'Safety Reporting in Clinical Trials' (planned for the third quarter)
- 'PV in the Non-commercial Environment' (date TBD)
- 'Differences in International Pharmacovigilance' (planned for the fourth quarter).

In addition, the Committee is currently working to update the 'Pharmacovigilance Auditing' booklet to reflect the new PV legislation. Publication is anticipated for May/June 2013. A benchmarking survey has been released to the membership, exploring how marketing authorisation holders are managing the pharmacovigilance system master file, now that the legislation is in effect. We hope as many of you as possible will participate to make this a truly worthwhile exercise!

Furthermore, the Committee is taking an active role in organising the programme for the 2013 European Conference, which will take place in Bonn, Germany in September.
Please watch this space for further updates on these and other Committee initiatives!

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For other committee members please see the committee pages on the website
www.therqa.com



UPDATE

Animal Health

SVEN BUCKINGHAM
ANIMAL HEALTH COMMITTEE CHAIRMAN

Committee Activities

The Committee have had one conference call since the last edition of Quasar.

A seminar 'Animal Health Site Audits – A Practical Approach' was held at Stansted airport on 13th February and was a great success – see the article below produced by Jo McKelvie and Janice Sarasola for further details. My thanks go to all members of the Committee who worked hard to produce the material – slides, workshops and the comprehensive checklist that was provided to delegates.

In response to the advertisement for new members, we are pleased to welcome Donna Watson of Triveritas to the Committee. We still have vacancies, so if anyone else is interested in joining, please contact Karl Butler or myself.

The Committee is actively involved in sourcing speakers for the first European QA Conference in Bonn later this year, with Marci Murphy being involved on the Programme Committee.



'Animal Health Site Audits – A Practical Approach' Seminar Radisson Blu, Stansted Airport 13th February 2013

A one-day seminar was held by the Animal Health Committee covering the topic of practical aspects of animal health site audits for VICH GCP studies. The aim of the seminar was to provide the attendees with practical examples of how to audit a range of animal health study sites performed to VICH GCP from the initial planning of the audit and on-site audit visit through to the close out and discussion of any audit findings and finally the production of the audit report and subsequent follow-up. Sessions provided useful take home examples of audit agendas, report formats and checklists for the on-site audits.

The seminar was well attended by animal health professionals with a range of experience in quality assurance, monitoring and management of animal health GCP studies. The day was composed of a series of presentations followed by corresponding workshop sessions. The day started with the seminar principal, Jo McKelvie (EVITA Services) covering the topic of planning and preparing for an on-site audit. Planning starts by determining the scope of an audit which is tailored to the specific study site such as veterinary practices, farms site, laboratories, feed mill (manufacturing of investigational veterinary products (IVPs)/ control products (CPs) in animal feed), universities, contract research organisations (CROs), etc. There are advantages during audit preparation to review documentation prior to the on-site visit to make the most of the time spent at the study site. The list of documents expected to be reviewed was provided. Since many study sites audited by the animal health auditor may be governed by certain restrictions, there are practical aspects of visiting these sites to keep in mind. This ranges from clothing requirements e.g. biosecurity in laboratories, farms, etc., to use of photography to capture information or the copying policy for documentation at that facility. With many clinical studies being carried out throughout Europe, the study language and the translations available is important for the auditor to be aware of during the planning stages.

This presentation was followed by three breakout workshop groups that 'prepared' an audit agenda/checklist for auditing a university, feed mill and veterinary practice.

The second session of the day was presented by Janice Sarasola (Ondax Scientific) who looked at some of the challenging areas that the auditor may face when auditing clinical studies being conducted at veterinary farms and laboratories. The aim was to enable the auditor to develop a range of suitable questions for these situations.

The particular areas covered within the farm environment were IVP/CP, animal feed and water, site equipment and animal identification, sample collection and despatch. The auditor needs to understand what data is important for the safety and efficacy evaluation within the study. This will help focus on the flow of the audit activities during the visit. Since the IVP/CP administration is one of the most important topics, time should be spent carefully reviewing the systems in use.

The IVP/CP systems in place covered the auditing approach to be used when the IVP/CP are delivered in feed or drinking water. Both the function of any delivery system used for medicated feed and/or water must be covered in the scope of the audit together with the methods in place to determine the final dose concentrations of IVP/CP and the monitoring of the stability of the IVP/CP during the dosing period which may of course cover several weeks/months. Auditors certainly need to keep fit as many of these IVP/CP systems can be in obscure places!

The sample collection and delivery procedures is another challenging area for the auditor as they will need to have a good understanding of the type of samples being taken and the storage and transport procedures to be adopted to ensure viability/conservation of the sample type.

This is particularly important for example, for biological samples where further cultivation is required at the designated laboratory.

The next topic covered was the audit of the laboratory which also incorporated the reception and processing of samples together with the facility aspects of the audit. The following workshop sessions gave the participants the chance to put into practice the topics covered.

Sven Buckingham (Buckingham QA Consultancy Ltd) covered the topic of the audit close-out including those important tips on conduct and dos and don'ts. Soft skills are important for an auditor to effectively communicate the findings and their impact. QA must maintain their independence while providing input into the corrective action preventative action (CAPA) process. The participants' skills were put to the test in the following workshop where they were presented with a range of thought-provoking scenarios in which the possible corrective actions and discussions of how the observations would be presented during the closing meeting were covered.

Marci Murphy (Boehringer Ingelheim Animal Health GmbH) presented on the reporting and follow-up of the site audit. An interesting topic on which I´m sure even the presenters learned a few new tricks of the trade! Excellent advice on keeping the report concise, complete and clear, especially when the report may be written in English which is not everyone's first language. The importance of the dreaded comma, what a difference that can make! There was an interesting discussion on classification of audit findings and how to evaluate this based on risk to the study. The report template examples given, together with tips for successful close out and follow-up, ended the presentation nicely.

The final workshop session gave us all the opportunity to have a go at concise report writing covering the important areas including:

- Description of the observation including evidence
- The standard which the observation is in non-compliance
- The non-compliance statement
- Risk classification.

This last one is perhaps something that many of us struggle with and the seminar provided an excellent forum for discussion.

The day was rounded up by Jo McKelvie and attendees were issued with the all important animal health site audit checklist, which was compiled by the Animal Health Committee to cover the range of questions that the animal health auditor can refer to when auditing:

- Personnel: sponsor/CRO, monitor, investigator
- Quality systems
- Study specific documentation
- Facility
- Animals
- IVP/CP
- Samples
- Computer systems
- Laboratory
- Others for consideration.

Feedback was very positive from all those who attended and certainly from the presenters point of view, a very enjoyable and action-packed day.

In the animal health arena the variety of site audits that an auditor may be expected to cover is perhaps far wider than that in other disciplines and as such, the preparation and training of the auditor needs to reflect those areas.

The Animal Health Committee are happy to provide further training on animal health related topics so please contact us with any good ideas. Many thanks to all those who attended and participated.

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UPDATE

Publications

CLAIRE VAN LEERSUM
PUBLICATIONS COMMITTEE SECRETARY

PUBLICATIONS COMMITTEE CONTACTS

Please See Editor's Page For Contact Details.

Articles on the Website

No need to wait for the Quasar submission dates! We have launched a new area on the website which means there is now the exciting opportunity for members to submit articles for publication on the website. This means that articles can be published outside of the deadlines for Quasar. The same rules and guidance applies for these as for the ones published in Quasar. Look out for the first one of these, the extended article featured on page 14 'Summary Comparison of Global GLP Regulations Pertaining to Liquid Chromatography – Mass Spectrometry Instrument Software Validation'.

New Feature in Quasar – Technical Toolkit

We are sometimes asked, especially in QA, to work in unfamiliar areas. A new Quasar feature called 'Technical Toolkit' aims to provide background information and key points. An article can't make us experts, but can address 'unknown unknowns' (also known as unconscious incompetence) and more. This should be useful for members in various situations, e.g. for an auditor as part of their preparation when auditing in a specialised area that they are unfamiliar with.

Please contact the Committee if you would like to write an article on your speciality. Some topics that have generated high interest include electrocardiograms, clinical emergency equipment/processes, vaccine science, computerised system validation of radiofrequency systems and infectious diseases. The topics are GCP flavoured so new and varied topics of interest are welcome. It isn't necessary for technical articles to address the edition theme.

RSS Feeds – Update Required

If you previously subscribed to the BARQA RSS feeds, you will now need to update your links to the new RQA RSS feeds.

Please visit
www.therqa.com/rss-feeds
for the current links.

Booklets

As promised the RQA booklets are being reviewed and coming out with the new RQA logo, see the website for details.

New Committee Member

Welcome to our new committee member, Hannah Farrer who joined in February 2013. Hannah is currently Quality Co-ordinator and Clinical Trials Administrator at Gregory Fryer Associates (GFA), an expert Regulatory, Clinical Research and Quality Services consultancy to the healthcare industry worldwide.

Prior to working at GFA, Hannah worked as a senior QA auditor for Syne qua non Ltd, a CRO specialising in data management, EDC systems, statistics and medical writing. She has over four years' experience of GCP and other quality system regulations and guidelines.

Hannah has been a member of RQA for four years, as well as an active member of the RQA Anglia Regional Forum Committee since 2009.



Publications Committee Vacancies

The Publications Committee currently has a vacancy for a member interested in assisting to raise RQA's profile through various platforms including Quasar, the website and booklets. The Committee's remit also includes liasing with the Board and other committees regarding publications and to seek out, gather and publish information which serves the needs of the membership.

The Committee meets face to face three times and holds four teleconferences each year. There is also activity in the interim, commissioning and reviewing submitted articles and proof reading material for publication in Quasar or in booklet format. You may be asked to sit on a booklet production group to assist with drafting of material to meet agreed timelines for publication.

Applicants from all GxPs are welcome. Please send your CV, together with a covering e-mail stating the reasons for your interest in joining the Committee, to the Chairman and Secretary.

Chairman: Sarah Howard: **publications@therqa.com**
Secretary: Claire van Leersum: **claire@cvlqa.com**

2013 RAMZAN VISANJI PREMIER ARTICLE COMPETITION

WIN £1,000
RQA TRAINING VOUCHER

This competition is open to everyone and automatically includes those published in Quasar and on the website. Articles should be fresh, forward thinking, thought-provoking and relate to quality issues.

All entries will be judged by members of the RQA Publications Committee. The winner will be featured in the January 2014 edition of Quasar and on the RQA website.

Please forward your articles to:
editor@therqa.com

So get creative and submit your article now!

UPDATE

Regional Forum

North of England Regional Forum Summary, 26th September 2012

The North of England Regional Forum entitled: 'Quality Outside the Box' was held on Wednesday 26th September 2012 at Arco National Distribution Centre, Hull. A total of 23 attendees (including speakers) attended the forum. Members of the North of England Organising Committee, Nick Rodopoulos (ZigZag Associates Ltd), Brian Prince (Exponent International) and Cate Ovington née Gardner (Covance Laboratories) hosted the meeting. Special thanks need to be extended to the staff at Arco spearheaded by Kate Hunton, for helping with the logistics and the organisation of the forum. The day started with registration, the all-important tea and coffee and welcoming speech by Nick Rodopoulos.

First to take the stand was host Mark Daniels, Standards Manager at Arco Ltd, with his work on **'Slim line Manual 15 – 1'**. Mark, an Arco employee for 11 years, was the lead auditor against the three standards ISO 9001, 14001 and 18001 the company was working towards. He explained that in an industry that is governed by guidelines and instructions, employees find it difficult to remember everything in great detail. He pointed out that their company potentially added to the problem by requiring employees to follow the company's 5776 word quality manual. Mark identified that the quality manual could be revamped to be more concise, easier to remember and follow without losing any of its detail and maintain the same standard of quality. He then went on to introduce his new quality manual of only 174 words and gave examples of how that is possible.

Next up was Hilary Tinsley with her presentation **'Working with GLP in China'**. Hilary joined Covance Laboratories in March 1997 and was the Team Leader for the QA-Chemistry team at the Harrogate facility. In her presentation she gave an account of the pharmaceutical industry in China and how the industry is dominated by non-branded generic products. Hilary explained the Chinese work ethic and emphasised how important it is to respect hierarchy, status, personal business relationships and connections. She gave details of the regulatory requirements for setting up a GLP study in China and gave details of the roadblocks that need to be overcome in order to assure the quality of GLP work. Notably she mentioned that there is no official English translation of the Chinese GLPs and that several terms are not consistent with SI 1999 No 3106. Finally Hilary gave a brief history and work of the regulatory authority of China, the SFDA and Covance's history of work in China.

After the break Sabrina Malone gave a speech on **'Quality Standards in Food Safety and Stewardship'**. Sabrina has been a Managing Scientist and a Quality & Food Safety Manager in Exponent's Centre for Chemical Regulation and Food Safety, for her company Exponent International Ltd. She gave an overview of the UK and EU legislative framework and food safety standards. She described how the certification of products is adequately and correctly specified, and how food business operators are ensured to be competent to produce specified products. She explained that verification of the competence of the suppliers is done by audits and questionnaires and that there is an established routine monitoring system for product examination and analysis. She added that customer complaints are confirmed monitored and assessed.

Finally she gave examples of food safety management systems that are applied, more notably on the conduct of a hazard analysis, determining critical control points and limits and establishing corrective actions.

Next to take the stand was Tim Jameson, Ethical Compliance Manager at Covance Laboratories. Tim, who has worked for Covance since 1984, has held a Home Office personal licence for 25 years and presented the **'New Animals (Scientific Procedures) Act'**. In his presentation Tim described the act and explained how it was transposed in European and UK law. More notably he mentioned that the draft amendments to 'The 1986 Animal (Scientific procedures) Act' have been published and were scheduled to be put before parliament in autumn 2012. He mentioned that the anticipated changes will be new conditions on project licences; there will be additional mandatory roles, such as the training and competence officer and information officer and that there will be a simplification on personal licences where the establishments are expected to control authorisation to carry out procedures but the individual will not be linked to an establishment.

After lunch served by Arco Ltd, Nazrul Khan took his turn in delivering his speech on **'New EU Pharmacovigilance Legislation'**. Nazrul has recently joined ZigZag Associates and has over 12 years' experience within drug development, pharmacovigilance (PV) and GCP auditing, including GxP regulated IT systems and has worked for the MHRA. He gave a summary of some of the GVP modules/guidelines released from a QA perspective and gave an overview of the main changes. He gave examples of some key changes such as a detailed description of PV systems will be phased out and replaced by the requirement of a PV system master file (PSMF). He stated that there will be a requirement for all CAPAs to be included in the PMSF, and the new legislation would cover expedited reporting timelines for non-serious adverse events and submission of PSURs and describes the need for a formal quality system and quality plan.

The presentation **'Compliance Auditing Computerised Systems - Theory and Practice'** was the next topic to be presented by Chris Montgomery (Boehringer Ingelheim). Chris chaired the RQA Computing Committee for four years and is the course principal on the computerised systems auditing and validation RQA courses. Chris established the auditing objectives of computerised systems by urging auditors to understand the risk factors of a system and appreciate what 'in compliance' means for a computer system and seeing the whole picture. He then discussed how to assess the validated state of the system and establish that it is fit-for-purpose. Chris hinted at various pitfalls such as the inability to separate the quality system from the computer system and the inability to separate the process from the system functionality and gave examples on how to avoid them.

Finally, after the last break of the day, Amy Griffiths talked about **'Out of Manufacturing into R&D/Commercial - An ISO Approach'**. Amy, Senior Specialist of Global Systems Quality Assurance at Johnson & Johnson, also had experience in compliance management and had gained her qualification as a lead auditor for the International Register of Certificated Auditors (IRCA). She illustrated what a good quality management system is and gave an account on where the benefits were in adopting one in the commercial environment. To explain the point further she described a case study where the ISO standards were adopted in a pharmaceutical local operating company.

Nick Rodopoulos in his closing speech provided some final thoughts on 'what RQA is' and invited the audience in joining the North of England forum at next year's meeting.

NICK RODOPOULOS, BRIAN PRICE AND CATE OVINGTON

South East Asia (SEA) Regional Forum – 2013 Milestones and Strategies



JULIE LOKE, JUN LI, MR. FOO YANG TONG, MARK KUO, SUMITRA SACHIDANANDAN

The beginning of 2012 marked the launch of the SEA Regional Forum. After a busy first year and a successful regional forum in October 2012, core team members were committed to continuing the positive momentum in 2013. We met on 24th January 2013 to discuss our milestones and strategies for the year 2013. In this meeting, we were pleased to welcome Mark Kuo from GSK to our core team and his valuable experience and enthusiasm were applauded by the existing team members.

It seems that we are going to be even busier this year! The group are looking into having the second and third regional forums held respectively in April and July 2013. There is a real goal from us to run one RQA training course by November 2013 in Singapore.

The group aims to reach out geographically to Taiwan, Korea, Vietnam and Indonesia for membership and to recruit an additional core team member from Korea.

We hope to establish links with the regulatory agencies in the SEA region and have made a good start with HSA Singapore in 2012. In 2013, efforts will continue to ascertain such links with regulatory agencies from Malaysia, Thailand, Vietnam, Hong Kong and Indonesia.

SEA Regional Forum core team members, Jun Li and Julie Loke from Johnson & Johnson, and Mark Kuo (GSK) met up with regulators from the Singapore Health Science Authority (HSA), Foo Yang Tong (Director, Clinical Trial Branch, HSA) and Ms Sumitra Sachidanandan (Compliance Inspector, Clinical Trial Branch, HSA) on 11th January 2013.

The discussion and agenda items included topics that would further enhance collaboration and partnership to facilitate knowledge and experience sharing regarding regulations and requirements, and to improve the quality of conducting research in Singapore.

In Singapore, the Clinical Research Professional (CRP) forum is held on a quarterly basis and HSA would be invited, usually in the fourth quarter of each year, to present and share inspection observations and trending for each year. We also hope to ensure that there is no duplication of work and objectives between the CRP forum and the RQA e.g. target audience, presentation and sharing topics, etc.

HSA kindly agreed to participate in SEA regional forums in the future e.g. on a six monthly basis. They had expressed an interest in topics regarding quality systems and how they could improve quality in research work via monitoring, auditing activities and a special focus on implementing an inspection programme for medical devices.

HSA has provided insight and examples on the importance of ensuring consistency in interpretation of the regulations, implementation of policies and procedures by the stakeholders and also emphasised the importance of collaborations with other research forums e.g. CRP. A joint meeting between CRP and RQA was also discussed to further collaborate and widen the target audience, and also improve knowledge sharing between clinical research professionals e.g. clinical research associates, clinical research coordinators and QA professionals.

The meeting with HSA regulators provided a good kick-start for the year 2013 to further collaborate with regulators in the SEA region and to build a quality environment in clinical research and development.

The group always welcomes contribution from all members on forum activities. You do not need to be in the core team in order to get involved. Please do not hesitate to contact one of us if you would like to participate or present any good ideas or topics on forum activities:

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VACANCIES: North of England and Anglia

The North of England is interested in expanding the existing team.

If you are interested in taking part, please contact Nick Rodopoulos

nick@zigzag.eu.com
+44 (0)7827 335 472

Hannah Farrer from the Anglia Regional Forum Group will be going on maternity leave from the end of April 2013.

If you are interested in joining the group as maternity cover, please contact Barbara Brown

barbara.a.brown@gsk.com

Anglia News

Next Forum: Hot Topics – we are delighted to inform you that the Anglia Regional Group will be holding its annual forum on 11th June 2013.

It will be held in the very popular and pleasant surroundings of the British Racing School in Newmarket and priced at the discounted rate of £65 for RQA members and £75 for non-RQA members. There will be a variety of speakers and will include updates on the various GxPs.

Booking is now available on the website.

Scotland and English Borders News

The good news is that we have recruited Donna Cunningham from CRL to bolster our numbers to keep the momentum going.

The next forum will be in September 2013 at a venue to be announced in due course. The proposed theme will be validation and calibration.

Australasian News

SAVE THE DATE

The next ANZ regional forum seminar is scheduled for Friday 23rd August, 2013 in Sydney.

If you have an idea or an offer to present, please contact Andrew King.

Process Mapping Course

Course is planned for 2nd to 3rd May 2013.
Bookings may be made via the website.

News: Australia New Zealand Therapeutic Products Agency

The governments of Australia and New Zealand have reaffirmed their commitment to establish a joint regulatory scheme. Developments can be monitored via the website:

www.anztpa.org

London and South East News

Next Forum: Quality, Cost, Time

Every project has the elements of quality, time and cost within it. These three fight each other as the project progresses and the project manager needs to keep each in control – but what matters most?

Date: Monday 20th May 2013

Venue: UCL, London

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- An Introduction to Managing an Audit Programme
- An Introduction to Quality Risk Management
- An Introduction to Risk-Based Quality Systems
- An Introduction to the UK Clinical Trials Regulations

Visit **www.therqa.com** for more information and to purchase the courses.

RQA Research Quality Association

Diary Dates



Professional Development Courses

April

Research Quality Assurance for Good Laboratory Practice
23rd – 24th April 2013

Practical Application of Quality Risk Management Tools and Techniques
23rd – 24th April 2013

May

Practical Pharmacovigilance Auditing
13th – 15th May 2013

GLP for Study Directors, Principal Investigators, Study Staff and Management
21st – 22nd May 2013

Systems Audit
21st – 22nd May 2013

June

Auditing Computerised Systems
3rd – 5th June 2013

Good Manufacturing Practice for Investigational Medicinal Products
4th – 5th June 2013

Process Mapping for Managers and Auditors
11th - 12th June 213

The Auditing Course
11th – 12th June 2013

Good Clinical Practice Regulatory Inspections
18th - 19th June 2013

Good Clinical Practice Auditing - Principles and Practice
25th – 26th June 2013

July

GLP for Study Directors, Principal Investigators, Study Staff and Management
2nd – 3rd July 2013

Regional Forums

London and South East Regional Forum
UCL, London
20th May 2013

Anglia Regional Forum
British Racing School, Newmarket
11th June 2013

All courses are held at Madingley Hall, Cambridge unless otherwise specified.
For full details of courses and seminars, see the RQA Professional Development Prospectus or visit the website.
All courses, seminars and regional forums are now available to book online.

www.therqa.com

Submit Articles For QUASAR

If you would like to submit an article on any of the future themes or for a topic you feel would be of interest, please contact the Editor:
editor@therqa.com

Next Issue:

July 2013 – Non-Regulated Research

Future Issues:

October 2013 – Audit Challenges
January 2014 – The Third International Edition
April 2014 – Change and Impact